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Empowering AI data scientists using a multi-agent LLM framework with self-evolving capabilities for autonomous, tool-aware biomedical data analyses

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Artificial intelligence agents are emerging as powerful applications of large language models (LLMs), automating complex tasks and enabling scientific data exploration. However, their use in biomedical data analysis remains limited by the difficulty of handling specialized tools and multistep reasoning. Here we introduce BioMedAgent, a self-evolving LLM multi-agent framework, which learns to use diverse bioinformatics tools and chain them into executable workflows through interactive exploration and memory retrieval algorithms. It allows biomedical users to initiate tasks using natural language, without requiring computational expertise. Evaluated on our newly released BioMed-AQA benchmark comprising 327 biomedical data tasks, BioMedAgent achieved a 77% success rate, surpassing other LLM agents, and generalized robustly to the external BixBench dataset. Beyond benchmarks, it autonomously performs cross-omics analysis, machine-learning modelling and pathology image segmentation, highlighting its potential to advance biomedical research and extend to other scientific domains requiring complex tool integration and multistep reasoning.

The biomedical field is experiencing rapid data expansion, driven by emerging technologies that generate vast amounts of medical texts, images and omics data[1,2]. The widespread adoption of electronic health records globally has greatly facilitated the collection of clinical data on diagnostics, treatments and follow-ups[3,4]. Advancements in digital pathology and high-resolution imaging technologies, such as whole-slide imaging, magnetic resonance imaging and computed tomography, have revolutionized diagnostic practices, ensuring greater accuracy, reproducibility and standardization[5–7]. Meanwhile, high-throughput multi-omics data, such as genomics, transcriptomics and proteomics, are accelerating our understanding of molecular functions in various physiological and pathological conditions[8,9] and driving advancements in disease research and drug development[10,11].

Analysing these large-scale biomedical data to drive scientific discoveries requires complex computational methods that integrate skills from bioinformatics, artificial intelligence (AI), software programming, statistics and mathematics[12]. These analyses are crucial for deciphering intricate genetic and molecular interactions, revealing biological mechanisms and disease origins[13–15]. Advanced machine learning and deep learning technologies further enable the discovery of

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novel biomarkers. AI-driven image interpretation, treatment response prediction and drug design highlight the role of computational technologies in transforming biomedical data into actionable knowledge and enhancing clinical decision-making[16–19].

To support these analytical needs, numerous computational tools have been developed for specific biomedical data tasks[12,20,21]. When integrated, these tools form complex workflows capable of processing diverse data types and producing notable results[12,22]. Platforms like Galaxy[23], Nextflow[22], Seven Bridges[24] and DNAnexus[25] aim to simplify workflow development and optimize the usage of these tools[12]. They make it easier for noncomputational users to assemble and execute workflows through user-friendly actions like clicking and dragging. However, these platforms still pose challenges, including dependency on predefined workflows that may reduce flexibility, outputs restricted to data files without interpretable summaries, and a lack of support for natural-language interaction, which restricts intuitive engagement.

Large language models (LLMs) represent a transformative step towards general AI by substantially enhancing natural-language understanding[26,27]. LLMs have shown remarkable potential in interpreting and reasoning, approaching human-level comprehension[28–31]. This has led to the development of autonomous agents capable of understanding their environment, making decisions and taking actions accordingly[32]. Rapid progress in LLM-based agents has enabled them to understand and generate human-like instructions, profoundly shaping scientific research[33], inspiring the concept of 'AI scientists'[34]. In the field of autonomous chemical studies, LLM agents have shown impressive performance for semi-autonomous experimental design and execution[35], as well as powering autonomous mobile robots for exploratory synthetic chemistry[36].

The language capabilities and autonomous analysis abilities of LLM agents make them promising solutions to overcome the limitations of existing workflow platforms. For example, a GPT-4-powered agent was used to annotate cell types in single-cell RNA sequencing (scRNA-seq) analysis[37]. BioChatter was designed to develop custom biomedical research software following open science principles[38]. ChatGPT Advanced Data Analysis (ADA) streamlined automated machine learning for clinical studies[39]. AutoBA, an AI agent, was designed for fully automated multi-omic analyses[40]. CellAgent, a multi-agent framework powered by LLMs, automated processing and execution of scRNA-seq data analysis tasks[41]. GenoTEX offered annotated code and results to solve gene identification problems[42]. These innovative applications of LLM agents highlight their potential to assist in biomedical data analysis.

Despite these advancements, current LLM agents for biomedical data analysis are constrained to specific tasks and data types, limiting their broader application. They struggle to effectively use diverse bioinformatics tools or chain them into workflows, making it challenging to plan and execute complex, multistep analytical problems. Furthermore, they lack the ability to efficiently reuse past experiences. Evaluating these agents remains challenging due to the lack of well-established benchmarks, with BixBench[43] being one of the few recently available. These limitations underscore the need for a more generalizable framework that can flexibly perform diverse biomedical data analyses and be systematically evaluated.

In this Article, we introduce BioMedAgent, a self-evolving LLM multi-agent framework, which learns to use various bioinformatics tools and chain them into executable workflows for carrying out diverse biomedical data tasks initiated by natural-language prompts. Given data and a scientific hypothesis, BioMedAgent functions as a virtual data scientist, performing multistep planning, autonomously exploring execution paths and summarizing final conclusions. Its tool-aware flexibility, multi-agent interactive explorations (IEs) and self-evolving capacity enable robust performance across diverse biomedical data tasks when systematically evaluated on our newly constructed benchmark and an external benchmark.

Results

LLM-powered multi-agent framework

BioMedAgent is a multi-agent collaborative framework that uses LLMs such as GPT, DeepSeek and Gemini to generate diverse agents. These agents engage in IE through a three-phase collaborative process: planning, coding and execution (Fig. 1a). During the planning phase, multiple agents collaborate to interpret user requirements, select appropriate tools and design an effective analysis workflow. In the coding phase, agents generate the necessary code to implement each step of the workflow, using local tools or web application programming interfaces (APIs). The execution phase runs these codes in each step and produces results.

To enable self-evolving capabilities, BioMedAgent incorporates the memory retrieval (MR) algorithm, which uses a retrieval-augmented mechanism to facilitate the reuse of successful experiences. It systematically records effective tool selections, workflows and executed codes, reusing and updating these memories with new insights as the system encounters similar tasks (Fig. 1a). This iterative cycle enables BioMedAgent to dynamically adapt and evolve over time, consistently enhancing its problem-solving capabilities.

BioMedAgent supports natural-language input, enabling users to interact with the system using simple text instructions rather than structured commands. For example, a user can simply input: 'Identify pathogenic variants on data <reads1.fq.gz>, <read2.fq.gz>.' BioMedAgent autonomously interprets the input and designs and executes the corresponding analysis workflows (Fig. 1a). This enables biomedical experts with no specialized computational or bioinformatics training to participate directly in their data analysis and interpretation, through natural-language interaction.

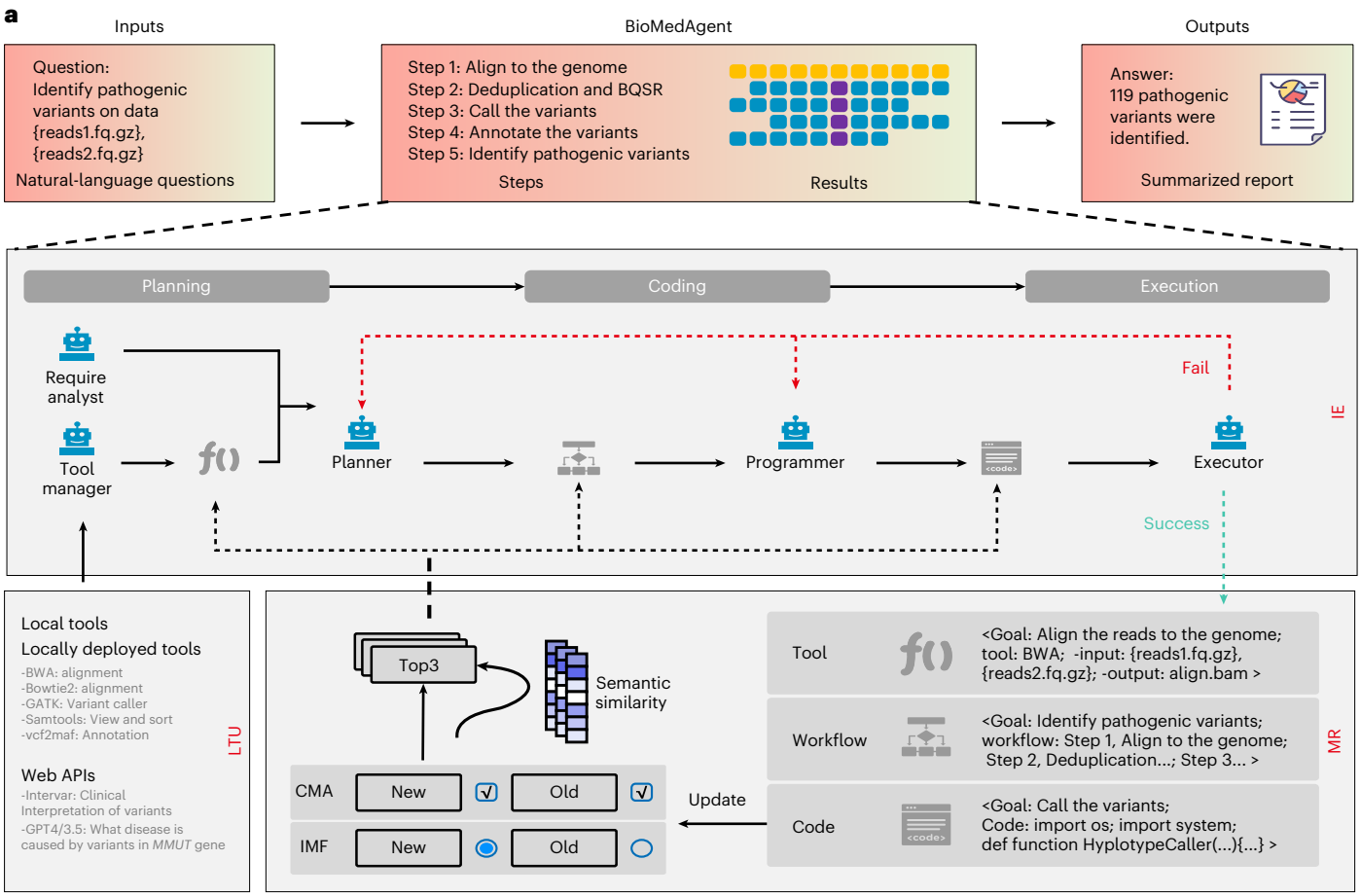
BioMedAgent's strong performance across diverse tasks relies on its ability to interpret user instructions and activate the appropriate tools. This is accomplished by the Tool Manager agent, which manages both locally deployed tools and web APIs within its workspace (Fig. 1a). These tools are crucial for specialized bioinformatics analyses, particularly in omics and precision medicine[44]. While the Programmer agent, derived from foundational LLM models, possesses general coding ability, it struggles to generate de novo implementations of specialized tools[45]. The Tool Manager agent addresses this by enabling LLMs to learn how to use existing tools from documentation covering descriptions, inputs, outputs and dependencies rather than rewriting new codes. Once integrated, these tools become available for autonomous use during task execution.

BioMed-AQA: benchmark of biomedical data tasks

To comprehensively evaluate the performance of BioMedAgent, we developed BioMed-AQA, a benchmark that includes 327 biomedical data analyses questions across five categories: omics analysis (O), precision medicine support analysis (P), machine learning (M), statistical analysis (S) and data visualization (V) (Fig. 1b). This benchmark is based on 17 types of biomedical data analysis tasks, including DNA, RNA, scRNA, classification, regression, deep learning and others. These diverse task types were selected to comprehensively assess BioMedAgent's performance across a wide range of biomedical analysis contexts.

We categorized the instructive questions in BioMed-AQA into clear-step and open-step types to test BioMedAgent's ability to translate natural-language inputs into actionable steps. Clear-step questions are those with well-defined steps (for example, 'Align sequencing data <tumor.R1.fastq.gz, tumor.R2.fastq.gz, normal.R1.fastq.gz, normal.R2.fastq.gz> using BWA, call somatic mutations with GATK's Mutect2, annotate variants with vcf2maf, and recommend therapy based on the genomic variants.'). Open-step questions are those in which only the objective is introduced, and the tools or steps are not clearly specified (for example, 'Perform mutation analysis on sequencing data <tumor.R1.fastq.gz, tumor.R2.fastq.gz, normal.R1.fastq.gz, normal.R2.fastq.gz> from cancer patients and recommend therapy.').

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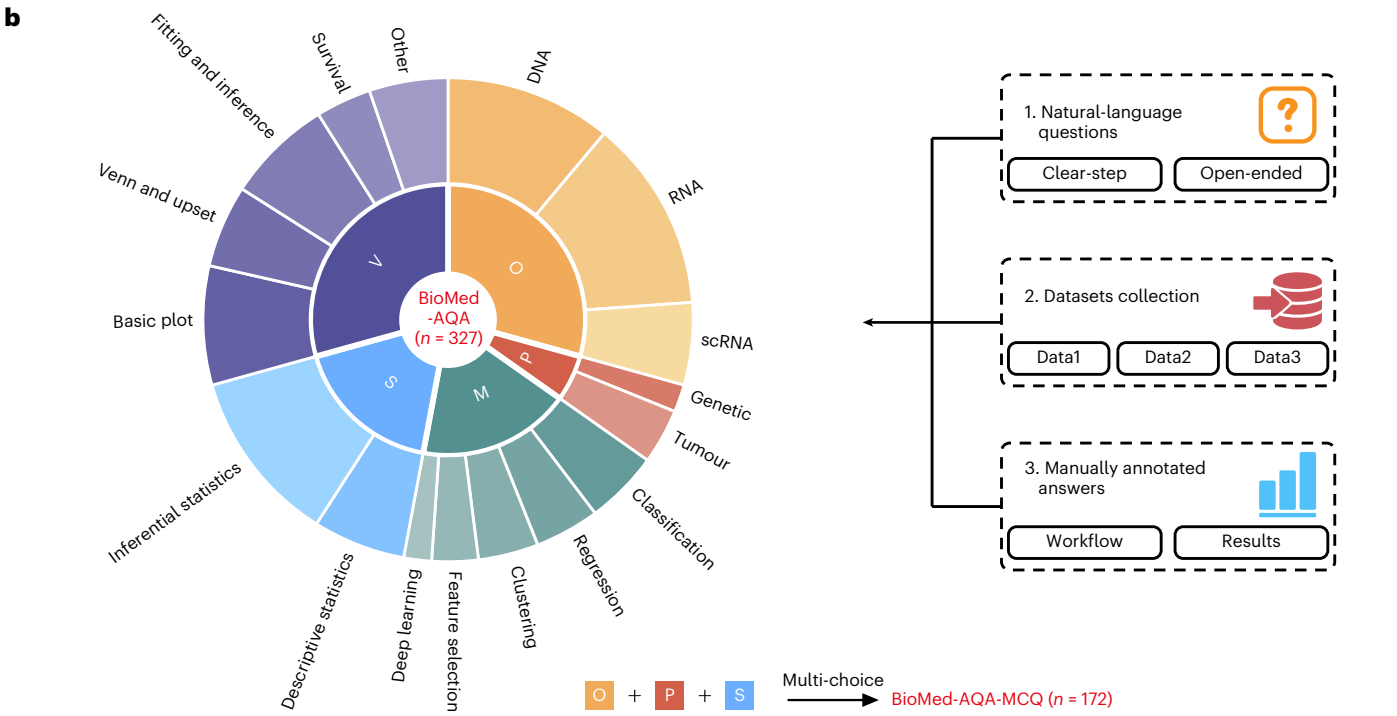


Fig. 1 | Framework and benchmark. a, Framework of BioMedAgent, from natural-language question inputs to analysis outputs, including three phases of planning, coding and execution supported by IE, MR and LTU. BQSR, base quality score recalibration. **b**, BioMed-AQA benchmark, comprising $n = 327$ manually curated biomedical data analysis questions and annotated answers across five categories: O ($n = 96$), P ($n = 18$), M ($n = 59$), S ($n = 58$) and V ($n = 96$). BioMed-AQA-MCQ, a subset of MCQs ($n = 172$) built from O, P, and S tasks.

Combined manual and automated evaluation using Win score

The effectiveness of BioMedAgent is assessed using a 'Win' score, designed to measure the system's ability to achieve predefined milestones (Extended Data Fig. 1a). Each of the 327 biomedical analysis questions is comprehensively annotated with the necessary steps and corresponding milestones for successful execution. A task that successfully meets all designated milestones receives a 'Win' score of 1, while a task that meets only one out of four milestones is scored as 0.25. A 'Win' score of 1 indicates an execution status of 'success', whereas a score below 1 is considered an execution status of 'failed'. The overall success rate is determined by calculating the percentage of tasks that receive a 'Win' score of 1.

Evaluating agentic AI is still a major challenge, especially for open-ended biomedical data tasks where there can be multiple correct ways to solve a problem. As observed in previous works such as AutoBA[40], BIA[46], BioMaster[47] and CellAgent[41], manual evaluation remains the most reliable approach, despite being labour intensive and limited in objectivity. In BioMed-AQA, manual assessment serves as the primary method, using the Win score above. This approach is necessary because the inherent uncertainty of generated content by agents often makes exact matching impractical—for example, differences in decimal precision (for example, '0.545' versus '0.55'), formatting issues (for example, '1/3' versus '1:3') or different visual representations (for example, survival curves created using different packages). Manual evaluation effectively addresses these challenges by considering contextual correctness rather than rigid exact matching.

To improve objectivity and scalability, we also introduced an autoscoring agent that compares the outputs against predefined milestones. It showed 92.3% consistency with manual evaluations (AUC = 0.926, $P < 0.001$; Extended Data Fig. 1b,c), reducing reliance on manual review. Furthermore, we also developed BioMed-AQA-MCQ, a 172-question multiple-choice questions (MCQs) subset, each containing five options, that supports fully automated, unambiguous evaluation (Fig. 1b and Extended Data Fig. 1d). Task from O, P and S categories in BioMed-AQA were supplemented with MCQs, while M and V tasks were not included, as they are less suitable for MCQ format.

Enhanced overall performance on O, P, M, S and V tasks

BioMedAgent has effectively shown its capability in handling general biomedical data analysis tasks, achieving an impressive overall average success rate of 77% across various categories on the BioMed-AQA benchmark (Fig. 2a). This includes high success rates in specific categories: 94% in O, 78% in P, 90% in M, 59% in S, and 65% in V (Fig. 2a,b).

We compared BioMedAgent's performance with other well-known LLM agents, all using the same underlying model, GPT-4omini. BioMedAgent achieved a superior success rate, with consistent improvements across O, P, M, S and V tasks, outperforming two of OpenAI's web agents and one local-tool-based agent. By contrast, the online ChatGPT-4omini was unable to provide data analysis, while the GPT Assistants achieved a 39% success rate, and GPT Function Call yielded a 33% success rate (Fig. 2a). Remarkably, it also outperformed the higher version model, ChatGPT-4o, which achieved a success rate of 46%, showing consistent improvements in the O, P, M and V tasks. By using local tools within its workspace, BioMedAgent showed a notably broader analytical scope compared to online ChatGPT-4o and GPT Assistants, achieving a 100% analysable scope. This exceeded the 68% and 65% analysable scopes observed for the latter two (Fig. 2a).

When tested with different open LLMs, including DeepSeek v3 and Qwen3, BioMedAgent maintained robustness, achieving 77% and 75% accuracy, respectively (Fig. 2a). The accuracy on MCQs was 76%, comparable to that on open questions (Fig. 2a). Although MCQs require completing the analysis before answering, they allow for some correct guesses, which explains the similar performance compared to open questions.

BioMedAgent's overall high performance remains consistent even as the complexity of workflows increases. Specifically, for P tasks, which typically involve an average of 6 planned steps, BioMedAgent continues to maintain a high success rate (Fig. 2a–c). While showing robust performance overall, BioMedAgent exhibited its lowest performance on S tasks. Although it surpassed other agent applications using ChatGPT-4omini, its success rate was lower than that of ChatGPT-4o, which uses a larger LLM model. This relatively weaker performance of BioMedAgent for S tasks may be attributed to the inherent demands of statistical tasks, which require fundamental data or table processing capabilities and thus rely more on the underlying LLMs.

Local tool usage significantly enhances the success rate

For biomedical data analysis tasks such as O and P, the use of specialized bioinformatics tools is crucial for conducting analysis effectively[44,48]. However, these tools often require complex environments and cannot be easily executed using standard Python packages, making it difficult for LLMs to program them correctly. To address this challenge, we enhanced BioMedAgent by establishing a Tool Manager agent responsible for learning and maintaining specialized tools with its workspace. This process involves integration of specific tools, which requires learning their documentation, including descriptions, inputs and outputs and preparing the necessary dependencies for execution. With this knowledge, BioMedAgent can activate the appropriate tools from the Tool Manager as needed.

We have integrated 67 specialized bioinformatics tools into BioMedAgent, mainly in the O, P and M categories. This integration has significantly improved BioMedAgent's performance, with average success rates for O, P and M tasks exceeding 90% (Fig. 2a,b). Moreover, tasks with local tool usage (LTU) show significantly higher Win scores compared to those without LTU across all tasks ($P = 1.259 \times 10^{-4}$, two-tailed t-test) and M tasks ($P = 1.477 \times 10^{-3}$, two-tailed t-test) (Fig. 2d and Extended Data Fig. 2a), with a similar trend observed in the success rate (Extended Data Fig. 2b). In cases where a required tool is missing, the agent can generate a custom tool code (CTC) to fill the gap. LTU and CTC complement each other, with LTU used exclusively in 46.25% of successful tasks, CTC in 28.85% and a combination of both in 24.90% (Extended Data Fig. 2c).

High robustness to varied prompts from human medical experts

Different users often phrase the same analytical tasks in their unique ways, influenced by their individual backgrounds and preferences. To evaluate how well BioMedAgent can handle these semantic variations, we conducted a test involving three human medical experts. Each human expert rephrased existing benchmark questions in more natural and varied ways without changing their analytical objectives. This allowed us to assess how well BioMedAgent understands the same task when expressed differently. The accuracy obtained from the three experts (0.761, 0.752 and 0.758) were comparable to the original benchmark accuracy (0.774) (Fig. 2e), indicating that BioMedAgent is robust to linguistic variations.

We further examined whether BioMedAgent can translate broad instructions into actionable analytical processes, by comparing its performance on clear-step and open-step questions that describe the same task with different levels of detail. Comparable Win scores were observed between the two types of questions across the total ($P = 0.622$, two-tailed t-test; Fig. 2f) and O, P, M, S and V tasks (Extended Data Fig. 2d); the success rates showed the same pattern (Extended Data Fig. 2e). The results suggested BioMedAgent's robust capability to autonomously plan and execute necessary processes, even in the absence of explicit step-by-step guidance. This adaptability highlights the system's advanced understanding, making it suitable for users without specialized computational or bioinformatics knowledge.

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Model	LLM used	Capable scope (total)	Success rate (total)	Success rate (O)	Success rate (P)	Success rate (M)	Success rate (S)	Success rate (V)
ChatGPT	GPT-4o	0.68	0.46	–	–	0.46	**0.62**	0.38
ChatGPT	GPT-4omini	0	–	–	–	–	–	–
GPT assistants	GPT-4omini	0.65	0.39	–	–	0.29	0.43	0.44
GPT function call (noIE)	GPT-4omini	1	0.26	0.43	0.39	0.37	0.03	0.15
GPT function call (IE)	GPT-4omini	1	0.33	0.55	0.83	0.37	0.05	0.15
BioMedAgent (noIE)	GPT-4omini	1	0.28	0.34	0.22	0.58	0.03	0.20
BioMedAgent (IE)	GPT-4omini	1	0.52	0.66	0.39	0.63	0.43	0.38
BioMedAgent (IE, MR rounds = 1)	GPT-4omini	1	0.70	0.88	0.72	0.86	0.53	0.52
BioMedAgent (IE, MR rounds = 2)	GPT-4omini	1	0.75	**0.97**	0.78	0.86	0.52	0.59
BioMedAgent (IE, MR rounds = 3)	GPT-4omini	1	**0.77**	0.94	0.78	0.90	0.59	**0.65**
BioMedAgent (IE, MR rounds = 2)	DeepSeek v3	1	**0.77**	0.96	0.67	**0.98**	0.53	0.63
BioMedAgent (IE, MR rounds = 3)	Qwen3	1	0.75	0.90	0.72	0.85	0.69	0.57
BioMedAgent (MCQ)	GPT-4omini	1	0.76	0.82	**0.94**	–	0.59	–

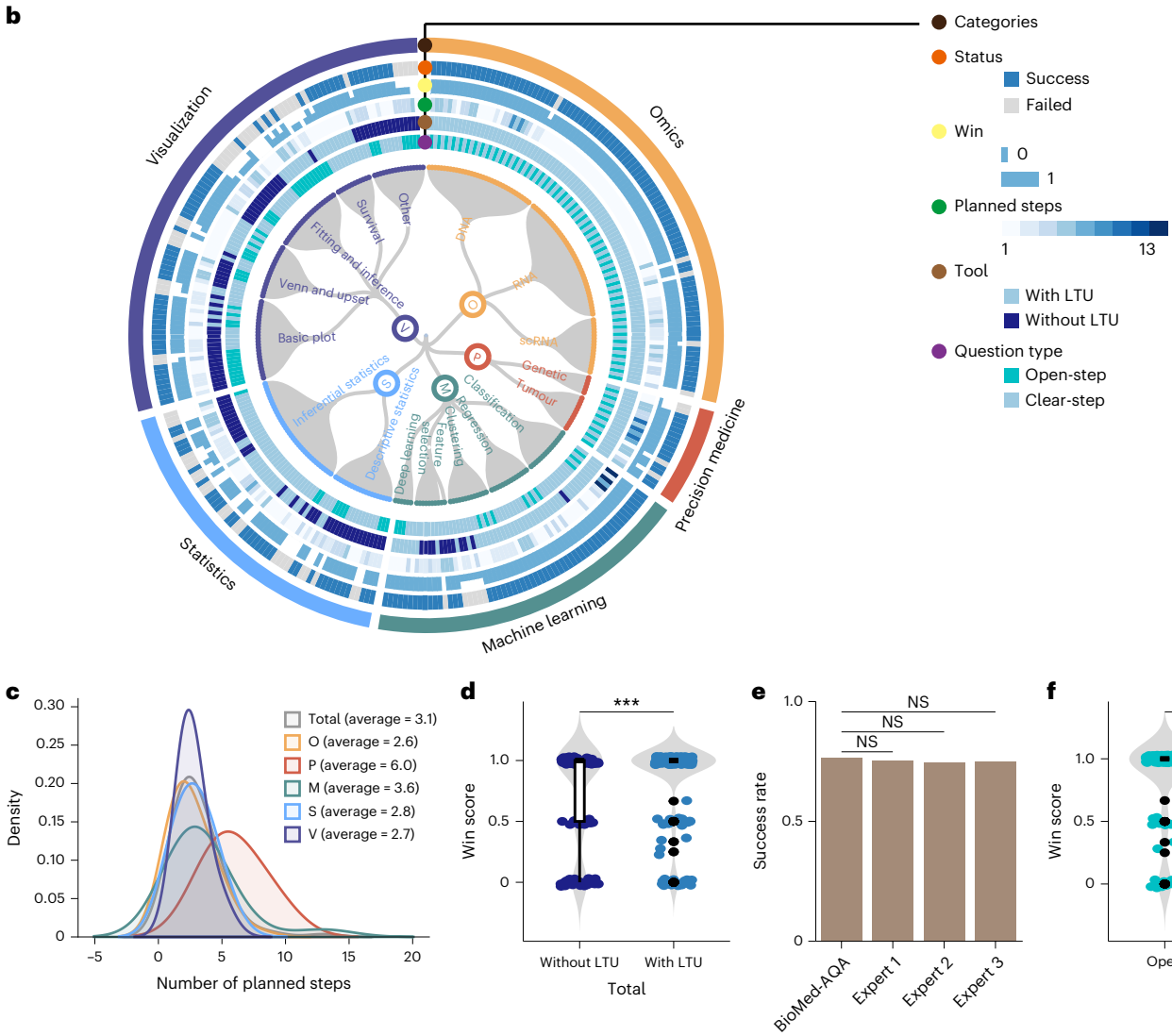


Fig. 2 | Performance of BioMedAgent. a, Comparison of BioMedAgent's performance with other LLM-based agents using different LLM across the five categories of O, P, M, S and V tasks. The last row shows results tested on MCQs ($n = 172$), while the others are from open questions ($n = 327$). Bold text indicates the best performance. **b**, Circos plot illustrating the performance of BioMedAgent after three rounds of learning. **c**, Number of planned steps by BioMedAgent across five categories of O, P, M, S and V tasks. **d**, Win score comparison between with and without LTU across total tasks ($P = 1.259 \times 10^{-4}$, two-tailed t-test; NS, $P > 0.05$; $^{*}P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$). **e**, Success rate comparison across varied human expert semantic instructions (two-sided chi-squared test). **f**, Win score comparison between open-step and clear-step questions across total tasks (two-tailed t-test).

Multi-agent collaboration enhanced by the IE algorithm

Human group intelligence involves collaborative explorations by multiple individuals to handle complex problems while using existing experiences to address similar problems[49]. BioMedAgent enhances the planning and execution of tasks by emulating human group collaboration, thereby increasing problem awareness and facilitating appropriate actions. While encountering problems for the first time, BioMedAgent activates IE to expand the explorations and optimize task planning and execution. While encountering problems again, it triggers MR to use the past experiences. In this way, IE uncovers potential solutions, while MR reduces redundant explorations (Fig. 1a).

BioMedAgent uses multi-agent collaboration enhanced by the IE algorithm to effectively address biomedical data analysis challenges (Extended Data Fig. 3a). This method has significantly improved the Win score, allowing more tasks to be analysed correctly. BioMedAgent with IE achieved the highest average Win score, outperforming noIE (without IE) strategy and surpassing two of OpenAI's web agents, as well as one local agent (Fig. 3a and Extended Data Fig. 3b,c). The average success rate nearly doubled, increasing from 28% to 52% when compared to noIE strategy ($P = 4.959 \times 10^{-10}$, two-sided chi-squared test; Fig. 3b). Consistent improvements were observed across five categories of tasks: O increased from 34% to 66% ($P = 3.818 \times 10^{-6}$, two-sided chi-squared test), P from 22% to 39% ($P = 0.469$), M from 58% to 63% ($P = 0.707$), S from 3% to 43% ($P = 1.341 \times 10^{-6}$) and V from 20% to 38% ($P = 0.011$) (Fig. 3b and Extended Data Fig. 3d).

As the number of IE rounds increased in both the planning and coding phases, the success rates consistently improved (Fig. 3c,d). IE expands the potential solution space and promotes innovations in group intelligence systems[49]. With the IE algorithm, the total number of explorations increased from 916 to 2,696, almost tripling, indicating that a significantly greater number of potential solutions were automatically explored ($P = 7.468 \times 10^{-15}$, two-tailed paired *t*-test) (Fig. 3e). Over 30.5% of successful tasks required extra interactions (rounds of IE ≥ 1) during the planning phase, while 48.6% involved extra interactions (rounds of IE ≥ 1) during the coding phase (Fig. 3f). This collaborative strategy enabled approximately 60.8% of previously failed tasks to succeed, highlighting the IE algorithm's effectiveness in boosting BioMedAgent's performance.

Self-evolving through the MR algorithm

The MR algorithm enables BioMedAgent to be self-evolving by leveraging a dynamic memory base built from previous tasks, allowing past experiences to be reapplied in new contexts (Fig. 4a and Extended Data Fig. 4a). When evaluated over three rounds of iterative learning using the BioMed-AQA benchmark, BioMedAgent showed progressive performance gains (Fig. 4b and Extended Data Fig. 4b), with the success rate rising from 52% to 77% (Fig. 4c). Across the five task categories, the success rate increased from 66% to 94% for O ($P = 3.103 \times 10^{-6}$, two-sided chi-squared test), 39% to 78% for P ($P = 0.043$), 63% to 90% for M ($P = 0.001$), 43% to 59% for S ($P = 0.137$) and 38% to 65% for V ($P = 3.070 \times 10^{-4}$) (Fig. 4c). BioMedAgent retrieved relevant past experiences through semantic similarity queries, in which analyses of the same type show closer semantic relationships, facilitating their application to related tasks (Extended Data Fig. 4c).

To ensure that the self-evolving did not result from direct memory access to identical questions and the subsequent recall of the answers, we further randomly reserved 30% of the BioMed-AQA benchmark as unseen questions and used the remaining 70% as seen questions. We performed iterative MR-based learning using only the seen subset and subsequently evaluated its performance on unseen subsets to see the system's generalization. The success rate showed a slight decrease for unseen questions compared to seen questions (69% versus 76%, two-sided chi-squared test, $P = 0.307$) yet remained significantly higher than before learning (69% versus 52%, two-sided chi-squared test, $P = 0.019$) (Extended Data Fig. 4d). These results confirm that the MR supports generalizable self-evolving, improving the agent's capabilities in tool selection, planning and coding, rather than relying on direct recall of previously generated answers.

Effective memory update strategies enable memory records to be optimally queried and applied in different analytical contexts[50]. BioMedAgent has established two distinct memory update strategies: continuous memory accumulation (CMA), which retains all successful records, and iterative memory forgetting (IMF), which selectively prunes outdated records (Fig. 4a and Extended Data Fig. 4a). IMF achieved faster convergence and slightly superior final success rates compared to CMA across all task categories (Fig. 4c and Extended Data Fig. 4e). It also showed greater stability in iterative rounds of learning, whereas CMA showed an abnormal decline in performance between the first and second rounds (Fig. 4c and Extended Data Fig. 4e). Furthermore, IMF required less intermediate memory, making it more economical in the long-term operation and maintenance of an expanding platform (Fig. 4c). Both IMF and CMA memory update approaches could reduce redundant re-explorations ($P = 1.646 \times 10^{-7}$ between rounds 3 and 0 for IMF, $P = 6.759 \times 10^{-8}$ for CMA, two-tailed paired *t*-test) (Fig. 4d).

After three rounds of learning with IMF, BioMedAgent achieved high performance across BioMed-AQA (Fig. 2a,b and Extended Data Figs. 5–8), showing its self-evolving capability for continuous improvement.

Robust generalization on external benchmark

Evaluating agentic AI is challenging, especially for biomedical data tasks where the same problem can be solved in several valid ways and the outputs cannot be judged by exact matching. There are few well-established benchmarks designed specifically for assessing scientific agentic AI. BixBench[43] is one of the few publicly available benchmarks, offering a comprehensive dataset over 50 real-world analytical scenarios ('capsules') and 296 associated questions. Its scale is comparable to BioMed-AQA, and it covers a wide range of bioinformatics tasks.

To evaluate BioMedAgent's generalizability beyond our curated BioMed-AQA benchmark, we tested it on BixBench, including both open questions and MCQs. When running without domain-specific tools and relying solely on CTC, BioMedAgent achieved 49% accuracy on open questions, 49% on MCQs with refusal and 63% on MCQs without refusal. These results significantly outperformed the benchmark-provided agent baselines of 37%, 42% and 55%, respectively ($P = 5.459 \times 10^{-5}$, 2.119×10^{-2}, 2.814×10^{-2}, two-sided McNemar's test) (Fig. 5a). These findings show BioMedAgent's strong capability in autonomous multistep biomedical analysis, even without tool integration, and highlight its robust generalization to external benchmarks.

Comparing with existing multi-omics AI agents

Several multi-omics AI agents have been recently developed, including BioChatter[38], BixBench[43], AutoBA[40], BIA[46], BioMaster[47], OLAF[51], ADA[39], GenoTEX[42], CellAgent[41] and CASSIA[52], each designed to solve specific biomedical data analytical tasks. To better illustrate the strengths of BioMedAgent, we conducted a structured comparison with these 10 existing multi-omics AI agents across three dimensions: functionality, extensibility and evaluation. Functionality is defined by the core capabilities such as planning, coding, execution and natural-language interaction. Extensibility refers to the ability to manipulate tools, support inter-agent communication and evolve over time. Evaluation focuses on the benchmark scale and assessment methods.

Most existing agents support the end-to-end process of planning, coding and execution but are often limited to specific analysis types (Fig. 5b). Among the more general agents, AutoBA provides support for omics analysis, while BioMedAgent shows broader applicability across O, P, M, S and V tasks. BioChatter also supports more general tasks by using knowledge graphs and live API services to enhance question-answering capabilities and to facilitate the development of biomedical software and databases within an open science framework.

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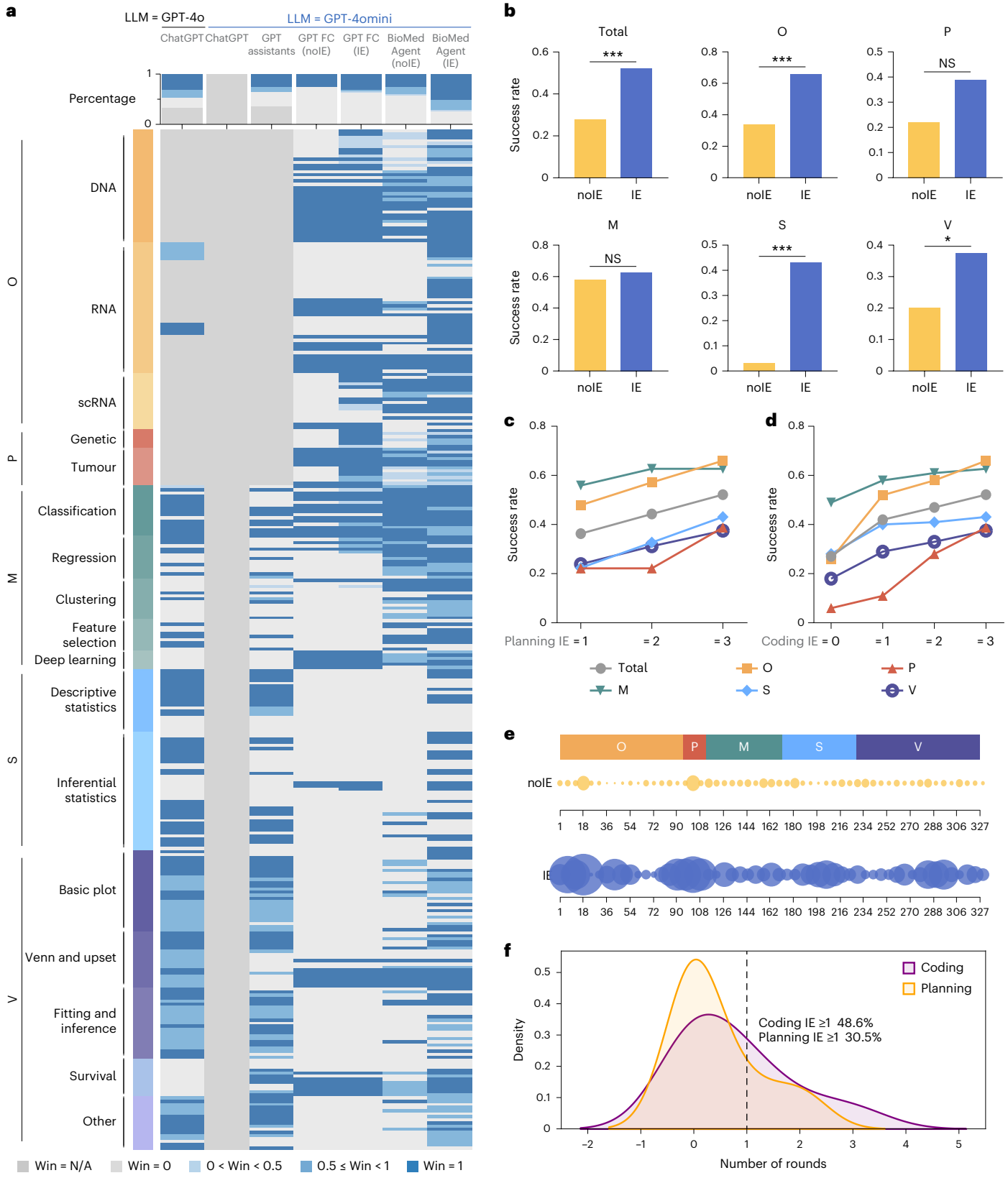


Fig. 3 | Evaluation of the IE algorithm. a, Heat map showing performance of different LLM-based agents across Total tasks of BioMed-AQA ($n = 327$). Colours represent different Win scores. GPT FC, GPT Function Call; N/A, not applicable. **b**, Success rate comparison for tasks with IE and without IE (noIE), with P values obtained via two-sided chi-squared test (total, $P = 4.959 \times 10^{-10}$; O, $P = 3.818 \times 10^{-6}$; P, $P = 0.469$; M, $P = 0.707$; S, $P = 1.341 \times 10^{-6}$; V, $P = 0.011$; NS, $P > 0.05$; *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$). **c**,**d**, The increase in IE rounds leads to improved success rates during the planning (**c**) and coding (**d**) phases. **e**, Counts of interactive explorations with IE and without IE (noIE). Circle size represents the count. **f**, Density plot of IE rounds during planning and coding phases for the tasks successfully executed.

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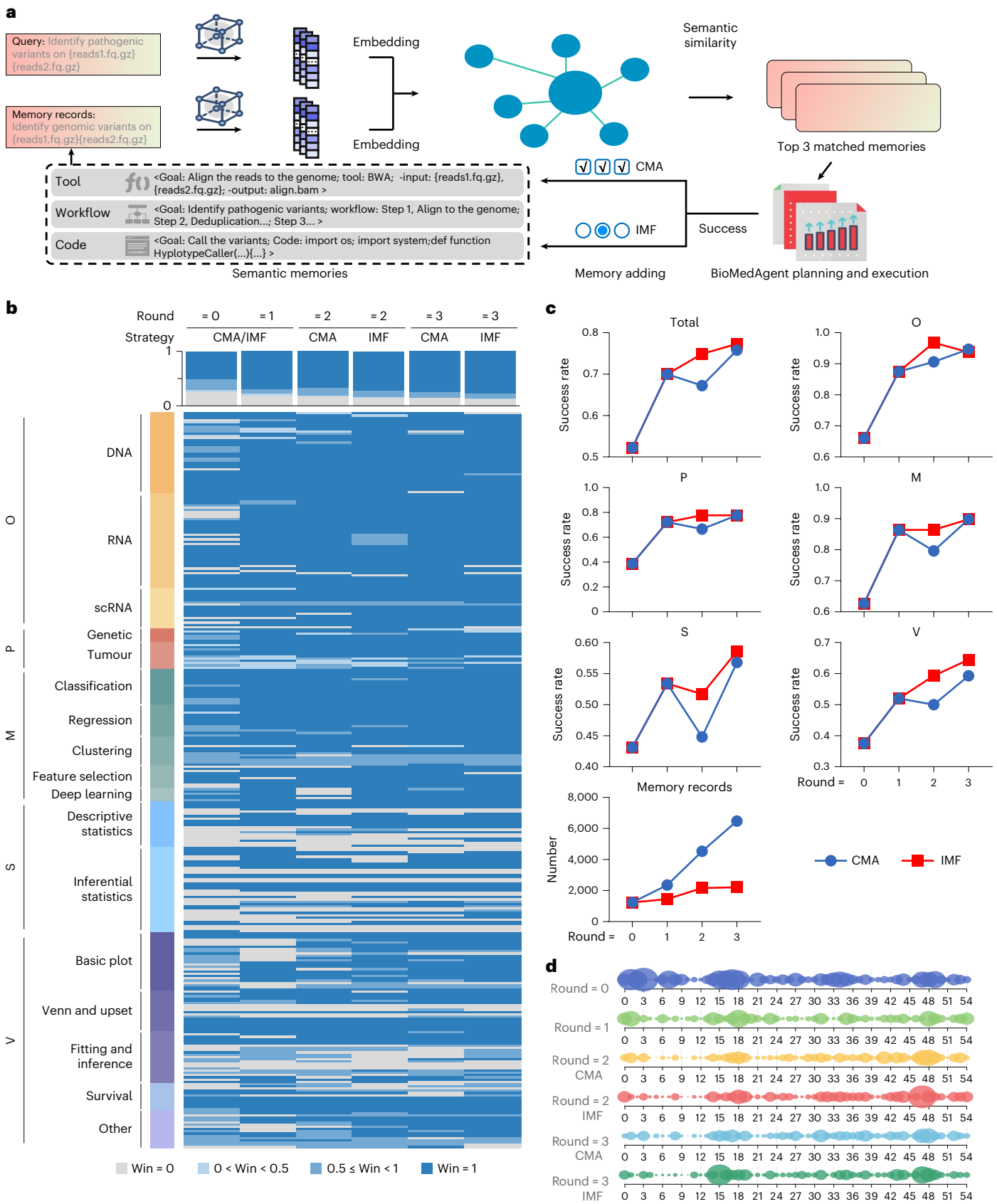
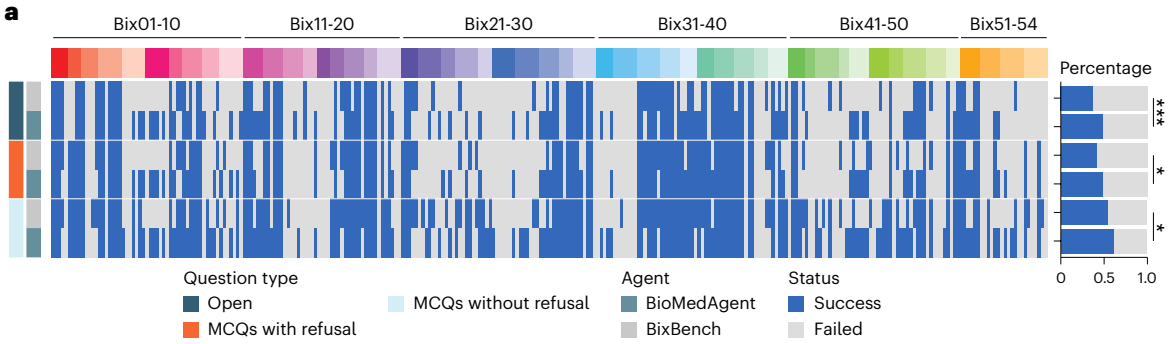


Fig. 4 | Self-evolving through the MR algorithm. a, The MR algorithm operates through a cyclic process involving memory formation, updating and reuse to continuously refine its capabilities. **b**, Heat map showing Win scores under different memory update strategies across Total tasks of BioMed-AQA ($n = 327$) in three rounds of learning. **c**, Comparison of success rate and number of memory records under CMA and IMF strategies in three rounds of learning. **d**, Counts of IEs under CMA and IMF strategies in three rounds of learning. Circle size represents the count.

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b

Agent	Functionality					Extensibility				Evaluation	
	Planning	Coding	Execution	NL instruction	Tasks	Local tool usage	Custom tool code	Multi-agent	Self-evolving	Benchmark scale	Evaluation Method
BioMedAgent	√	√	√	√	O/P/M/S/V	√	√	√	√	327 open / 172 MCQs	Manual / autoscoring
BioChatter	×	√	√	√	O/P/M/S/V	√	×	√	√	living bench	Autoscoring
BixBench	√	√	√	√	O/P/M/S/V	x	√	x	×	296 open / 296 MCQs	Autoscoring
AutoBA	√	√	√	√(half)	O	×	√	×	×	40 open	Manual
BIA	√	√	√	√	O	×	×	√	√	15 open	Manual
BioMaster	√	√	√	√	O	√	√	√	×	20 open	Manual
OLAF	√	√	√	√	O	x	x	x	×	1 open	Manual
ChatGPT ADA	√	√	√	√	M	×	√	×	×	5 open	Autoscoring
GenoTEX	×	√	√	√	M(key gene)	x	x	x	×	82 open	Autoscoring
CellAgent	√	√	√	√	O(scRNA)	√	√	√	√	17 open	Manual
CASSIA	×	×	√	×	O(scRNA)	√	×	√	×	5 datasets	Autoscoring

Fig. 5 | Evaluation on external benchmark BixBench and agent comparison. **a**, Evaluation of BioMedAgent on the external benchmark BixBench with over 50 real-world analytical scenarios (Bix01-54, $n = 296$ data analysis tasks), showing significantly higher accuracy compared to that of the BixBench-provided baseline agent across open questions, MCQs with refusal and MCQs without refusal ($P = 5.459 \times 10^{-5}$, 2.119×10^{-2}, 2.814×10^{-2}; two-sided McNemar's test; NS, $P > 0.05$; *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$). **b**, Comparison of BioMedAgent with other multi-omics AI agents, highlighting BioMedAgent's superior functionality, extensibility and evaluation coverage, including broader tasks, dynamic tool integration, adaptive evolution and large-scale benchmark support. √, supported; ×, not supported; open, open questions; MCQs, multiple-choice questions.

Extensibility remains limited in most agents, as few support dynamic tool integration and adaptive evolution. By contrast, BioMedAgent combines both LTU and CTC, reinforced by a self-evolving mechanism via the MR algorithm. Regarding evaluation, only BixBench and BioMedAgent offer large-scale benchmarks (300 tasks). Most others rely on small application demos. While open questions are the dominant evaluation method, only BixBench and BioMed-AQA include MCQs to support automated and objective evaluation. Moreover, most existing evaluations depend on manual annotation, whereas BixBench and BioMedAgent support autoscoring assessment.

In summary, BioMedAgent offers broader task coverage, greater flexibility in tool usage, adaptive evolution and a more systematic and scalable evaluation framework compared to existing agents.

Applying BioMedAgent to cross-omics analysis of RNA-seq and scRNA-seq data

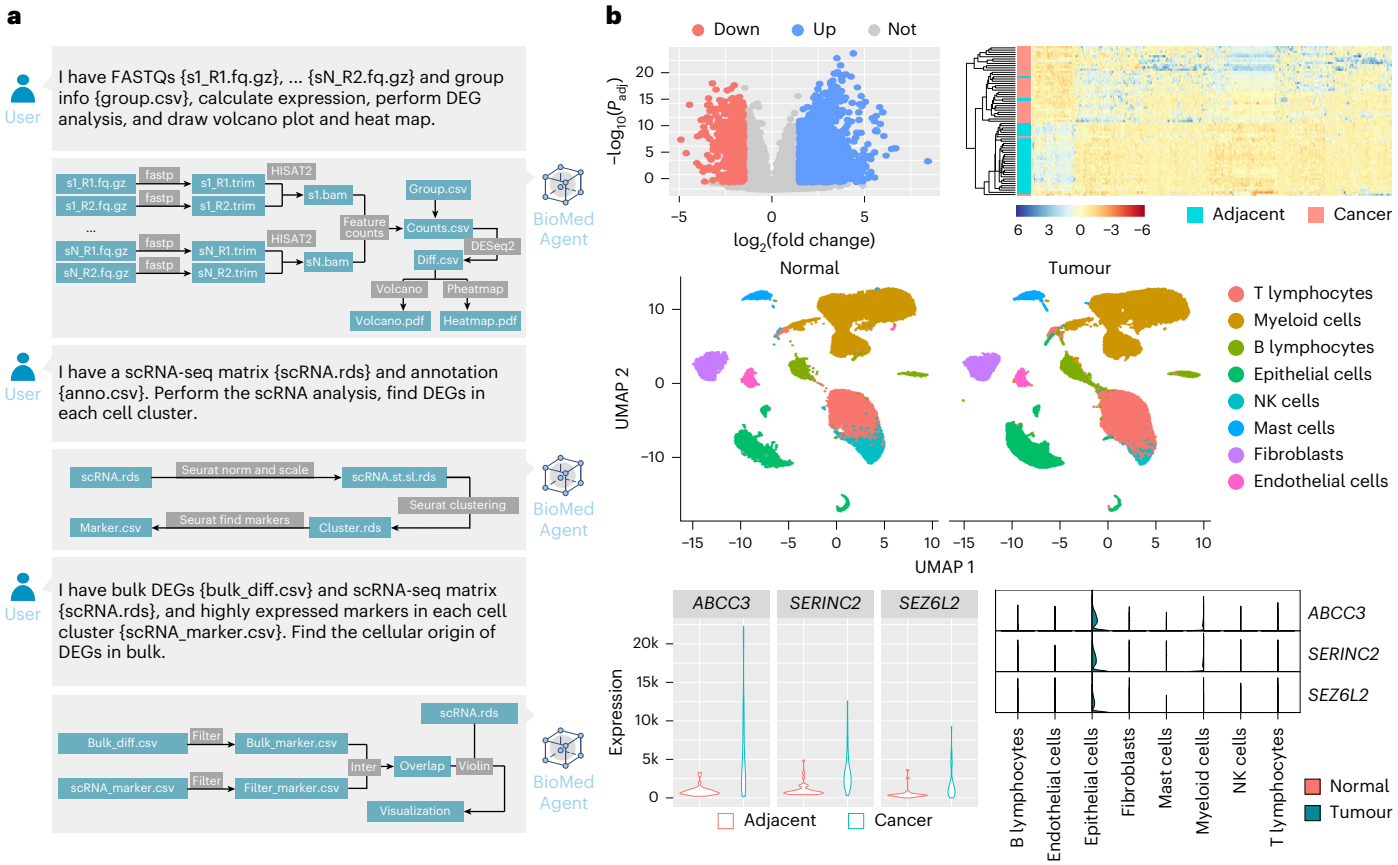
To assess BioMedAgent's ability to assist in complex biomedical research, we designed a cross-omics question: From which cell types are the highly differentially expressed genes (DEGs) in non-small cell lung cancer (NSCLC) derived? This requires integrating molecular and cellular information.

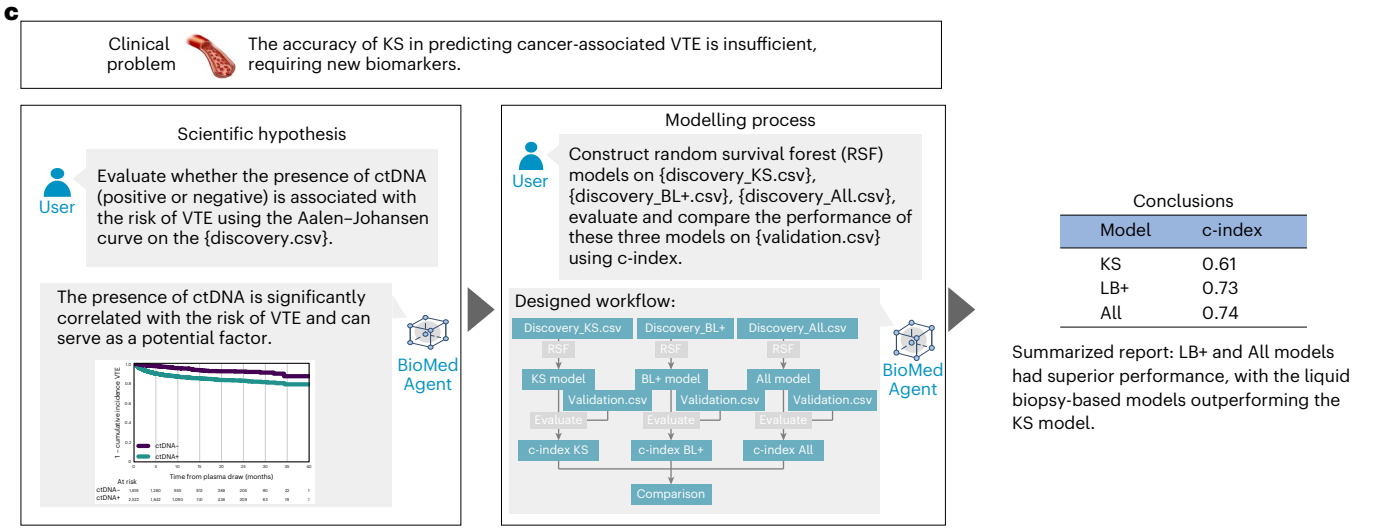
We downloaded bulk RNA-seq data from 67 NSCLC samples (34 tumour and 33 adjacent normal samples) and 22 scRNA-seq matrix files from the Gene Expression Omnibus (GEO, GSE268175, GSE131907). Using three natural-language instructions, BioMedAgent conducted large-scale cross-omics data analysis, covering bulk RNA-seq, scRNA-seq and cross-omics integrative mining. It automatically programmed and executed the planned steps, which included trimming, alignment, quantification, visualization for RNA-seq, normalization, scale, clustering, marker identification for scRNA-seq and filtering, intersection and visualization for cross-omics analysis (Fig. 6a).

A total of 1,831 DEGs, including 1,309 upregulated and 522 downregulated, were identified from bulk RNA-seq data, with corresponding volcano plot and heat map for visualization (Fig. 6b). These covered 78% ($n = 842$) of the DEGs identified by the GEO2R online tool, showing consistent findings (Extended Data Fig. 9a). Eight cell types were detected, and cell-specific DEGs were identified from the scRNA-seq data (Fig. 6b). Cross-omics integration revealed NSCLC-associated cell-specific DEGs: *ABCC3*, *SERINC2* and *SEZ6L2*, which were predominantly overexpressed in epithelial cells (Fig. 6b). *ABCC3* encodes a protein belonging to the MRP subfamily which is involved in multi-drug resistance and plays a key role in NSCLC prognosis[53]. *SERINC2* is associated with cervical cancer[54,55], and *SEZ6L2* is upregulated in lung cancer, which can be used for predicting the prognosis[56,57].

Using natural-language instructions, BioMedAgent autonomously conducted a large-scale cross-omics analysis and produced results

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d

Cell type	Baseline	BioMed Agent	Absolute Dice gain	Remaining gap closed
Inflam-matory	0.806	0.813	+0.87%	+44.60%
Epithelial	0.817	0.821	+0.49%	+35.10%
Spindle-shaped	0.824	0.842	+2.18%	+88.70%
Other	0.803	0.817	+1.74%	+57.90%
All	0.816	0.823	+0.86%	+29.90%

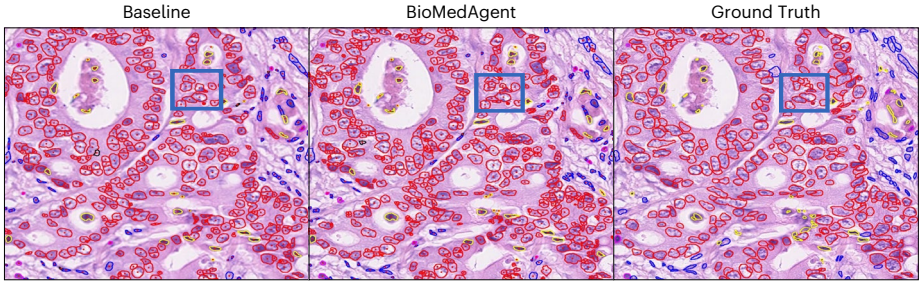


Fig. 6 | Applying BioMedAgent to accelerate diverse biomedical data research. **a**, Three natural-language instructions and the planned workflows by BioMedAgent to conduct large-scale cross-omics data analysis. **b**, Visualizations for the cross-omics data, including volcano plot and heat map for bulk RNA-seq, cell clustering for scRNA-seq and violin plots for cross-omics. UMAP, uniform manifold approximation and projection; NK, natural killer. **c**, Streamlined machine learning modelling for cancer-associated VTE prediction based on ctDNA data. On the left is the hypothesis, with replication of the modelling in the middle, while on the right is the evaluation of the models using c-index and the summarized report. **d**, Facilitate modelling of cell segmentation and classification with pathological image resolution enhancement. Comparison of Dice value and remaining gap closed for BioMedAgent and baseline. An example of the segmentation is shown on the right.

consistent with outputs from official online tools and with the published literatures.

Streamlined machine learning modelling for cancer-associated venous thromboembolism prediction

Machine learning is difficult for biomedical experts due to the complex requirements of data preprocessing, model selection and outcome evaluation. To test whether BioMedAgent can simplify this using natural language, we reproduced a *Nature Medicine* study[58] that built a liquid biopsy model using circulating tumour DNA (ctDNA) to assess the risk of venous thromboembolism (VTE) in cancer patients.

Using natural-language instructions, BioMedAgent automatically generated and executed a complete end-to-end machine learning modelling workflow for this task, encompassing model training, prediction, performance evaluation and comparison. It reproduced the entire analytical process described in the publication, from hypothesis formulation to model validation and interpretation, without manual coding. The results validated the hypothesis and confirmed the conclusions of the original study, showing that both the LB+ model (liquid biopsy plus two minimal clinical variables) and the All model (full model including all variables) showed superior predictive capabilities, with c-index values of 0.73 and 0.74, respectively (Fig. 6c). By contrast, the Khorana score model showed a lower predictive performance with a c-index of 0.61. These results showed that liquid biopsy-based models significantly outperform the Khorana score model in predicting VTE risk.

Overall, BioMedAgent enables biomedical researchers to run complex machine learning pipelines directly from natural language, bridging the gap between clinical expertise and computational technologies.

Improved cell segmentation and classification on pathological images

Cell segmentation and classification is crucial for pathological image analysis, as precise cell segmentation results can support downstream tasks such as lesion identification and survival assessments. However, cell segmentation tasks in pathology face challenges of low-resolution images. Recent deep learning algorithms can enhance resolution by transforming low-resolution images into high-resolution. We used BioMedAgent to integrate cell segmentation with resolution enhancement algorithms, aiming to enhance the effectiveness of cell segmentation. We guided the construction of an automated workflow by incorporating MiHATP v.1.0[59], a resolution enhancement tool, into BioMedAgent's local workspace. BioMedAgent's planned workflow automated data splitting, image resolution enhancement, cell segmentation and Dice evaluation (Extended Data Fig. 9b).

Compared to the baseline model, BioMedAgent showed consistent improvements in absolute Dice scores across four cell types: inflammatory, epithelial, spindle-shaped and others, with an overall Dice gain of +0.86% (Fig. 6d). In addition to the absolute gain, we also calculated the relative improvement compared to the remaining gap to the upper limit (referred to as 'Delta Dice to upper'). The results showed that the remaining gap closed reached 29.9% overall. These absolute and relative gains showed the practical value of integrating BioMedAgent with resolution enhancement algorithms, consistently improving segmentation accuracy across various cell types.

Local deployment and intuitive chat interface

BioMedAgent can run locally while calling online LLMs (GPT-4/4o/4omini, Claude 3) or local open-source models (for example, Qwen3-235b-a22b). It provides an intuitive chat interface that makes it easy to start tasks, monitor execution and view results, so complex workflows become manageable on a local set-up (Extended Data Fig. 10). This design is lightweight, easy to deploy, and scalable, making BioMedAgent usable for researchers with diverse technical backgrounds.

Discussion

BioMedAgent represents an important step toward autonomous, data-centric biomedical analysis by using a flexible multi-agent system architecture. Unlike conventional LLM-based approaches that primarily generate textual responses or isolated code fragments, BioMedAgent reconfigures LLMs into specialized, interacting agents capable of end-to-end data workflows. Rather than functioning as a question-answering system, it autonomously plans, codes, executes and adapts, thereby bridging the gap between AI automation and practical biomedical data research needs.

A core advantage of BioMedAgent lies in its tool-aware flexibility. Unlike general-purpose multi-agent frameworks such as LangGraph, CrewAI and SmolAgents, which rely on fixed agent roles and predefined tools, BioMedAgent supports dynamic integration of both LTU and CTC. By learning tool usage directly from documentation and adapting to task-specific requirements, the agent can incorporate validated domain-specific tools when available or generate custom code to address new or unsupported analytical steps. This dual mechanism ensures high robustness in established scenarios and strong adaptability when faced with novel tasks.

BioMedAgent's intelligence arises from its multi-agent and self-evolving design. Through the IE algorithm, agents iteratively exchange feedback and context across planning, coding and execution phases, enabling rapid, targeted explorations and adjustments. Each agent maintains a dedicated memory space, storing successful experiences and using the MR algorithm to recall and reuse the past experience. This continuous learning process enables BioMedAgent to improve autonomously over time.

An important design is the balance between exploration and experience reuse. Exploration can lead to new discoveries but consumes resources, while relying on past experience may limit innovation[49]. BioMedAgent currently adopts a stratified approach based on semantic similarity, automatically determining whether to try new explorations or use existing experiences, without manual tuning. In future iterations, introducing a user-controllable index could allow dynamic adjustment of this balance, providing finer control over computational cost and innovation.

Another consideration is the integration of both positive and negative experiences. Currently, BioMedAgent, like most LLM-based systems, preferentially records successful entries[60]. However, failure cases provide valuable insights for avoiding repeated mistakes and enhancing robustness[50]. Understanding and integrating these unsuccessful experiences is also meaningful, and future research aimed at more effective use of 'negative experiences' could be beneficial for BioMedAgent[61].

The efficiency of memory update mechanisms is critical for sustaining long-term self-evolving[62–64]. Our findings indicate that the IMF strategy yields faster and more effective adaptation than CMA, although the results are based on limited rounds and task sets. More advanced memory update strategies, such as mathematical scoring, could be further investigated as the system evolves and expands.

BioMedAgent still has limitations that are common in LLM-based systems, particularly hallucinations arising from next-token probabilistic modelling. In multistep workflows, it may fabricate intermediate results or final conclusions by failing to correctly track previously generated outputs. In addition, the performance of the agent system is inherently limited by the reliability of the specialized tools within its workspace, and scaling up tool numbers may introduce computational bottlenecks. Future work should focus on reducing hallucinations to enable trustworthy reasoning, developing a more standardized tool infrastructure and evolving from closed-loop digital interactions to more open real-world engagement.

In summary, BioMedAgent advances the vision of an 'AI data scientist', capable of autonomously conducting complex, data-centric biomedical research and continually improving through accumulated experience.

Methods

BioMed-AQA benchmark

The datasets in BioMed-AQA were manually curated from three sources: (1) simulated datasets, (2) tutorial examples from commonly used bioinformatics tools or packages and (3) literature-derived datasets from biomedical research. Among them, simulated datasets account for 37.3%, tool tutorial datasets for 15.9% and literature-derived datasets for 46.8%. Together, these datasets cover five categories of O, P, M, S and V tasks.

O and P task datasets. In the DNA-sequencing (DNA-seq) analysis, we used VarBen v.1.0 (ref. 65) to simulate DNA-seq data that include both germline and somatic tumour mutations. First, we downloaded whole exome sequencing data from the National Institute of Standards and Technology Genome in a Bottle consortium (NA12878)[66] and aligned it to the genome using the GATK v.4.0 Best Practices workflow to obtain alignment files in Binary Alignment Map format. We then used VarBen to feature pathogenic germline mutations on *MMUT*, *PAH* and *SLC22A5* genes and oncogenic somatic tumour mutations on *EGFR*, *KRAS* and *BRAF* genes, resulting in three germline and three tumour sequencing datasets, together with control samples. Precision medicine datasets were generated from these DNA-seq data to support genetic disease prediction and drug analysis in tumour samples.

For RNA-seq, we used three sets of expression profile data: one microarray dataset from the GEO database (GSE19089) and two next-generation sequencing datasets from the University of California, Santa Cruz Xena Browser (https://xenabrowser.net/), including The Cancer Genome Atlas Colon Adenocarcinoma and The Cancer Genome Atlas Rectum Adenocarcinoma. Cancer and normal control labels from the raw data were used as grouping information.

For scRNA-seq, we used three datasets: the first dataset is a peripheral blood mononuclear cell (PBMC) dataset provided by 10X Genomics and used in the Seurat tutorial[67], containing approximately 2,700 single cells sequenced on the Illumina NextSeq 500; the second dataset is oesophageal cancer data obtained from a published study[68]; the third dataset is another PBMC dataset provided by 10X Genomics, used in the Scanpy tutorial[69], containing approximately 68,000 single cells sequenced on the Illumina NextSeq 500.

M task datasets. Machine learning includes classification, regression and clustering tasks. For classification, three datasets from Kaggle (https://www.kaggle.com/datasets) were used: the Diabetes dataset, the Heart Attack Risk Prediction dataset, and the Predicting Heart Failure dataset. For clustering, the Breast Tissue dataset from the UC Irvine Machine Learning Repository (https://archive.ics.uci.edu/datasets) and the two Kaggle datasets (the Heart Attack Risk Prediction dataset and the Predicting Heart Failure dataset) were used, with classification labels removed for clustering. For regression, three datasets from Kaggle—the Medical Cost Personal Datasets, the Diabetes dataset and the Heart Attack Risk Prediction dataset—were used to predict medical expenses and the risk of disease occurrence.

S and V task datasets. For statistical analysis, the *t*-test task datasets were obtained from the demos in the R v.4.3.0 package for 'stats::t.test'. The Wilcoxon-test task datasets are from three built-in R datasets: airquality, birthwt and iris. The LEfSe task datasets were derived from our previously published microbiome data. The survival task datasets were from TCGA, including ovarian carcinoma, liver hepatocellular carcinoma and lung adenocarcinoma and include curated phenotype information such as stage, gender and *KRAS/EGFR/ALK* mutation status. For data visualization, the enrichment bubble plot task dataset was from the demos provided on the KOBAS v.3.0 website[70], and the Venn plot task datasets were from the official examples provided in the R package VennDiagram.

Evaluation design

Open questions and MCQs. The BioMed-AQA benchmark contains both open questions and MCQs to comprehensively evaluate BioMedAgent's performance in biomedical data analysis. The open questions ($n = 327$) are designed to reflect real-world bioinformatics scenarios, with annotated milestones provided as reference answers. By contrast, the BioMed-AQA-MCQ ($n = 172$) are more structured and represent a subset of BioMed-AQA, as M and V tasks are excluded because they are not suitable for a MCQ format. This subset includes 46 MCQs and 126 single-choice questions, each offering five answer options (Extended Data Fig. 1d). During testing, we did not indicate whether the question was single-choice or multiple-choice to minimize the likelihood of chance-correct responses.

Comparison against OpenAI agents. We conducted a comparative evaluation with three OpenAI agent frameworks: ChatGPT, Assistants API and Function Calling (using GPT-4o and GPT-4omini models). As OpenAI's underlying agent framework is closed-source, we relied on their official online services or API for this comparison. ChatGPT functions as OpenAI's public-facing interactive interface; we submitted tasks along with the input data through the chat window. Assistants API is part of OpenAI's hosted agent service, which supports file uploads to the user workspace; we used the API to upload input data, send analysis prompts in batches, and write and execute Python code through its Code Interpreter to iteratively handle complex analyses. Function Calling allows for invoking external tools by creating local APIs and exposing them to the agent through structured definitions; we tested its ability by using local tools (same as we used in BioMedAgents) provided in its environment. We used the identical task prompts and data inputs from BioMed-AQA to ensure consistency in comparison with BioMedAgent.

Evaluation on external benchmark BixBench. We tested BioMedAgent's performance on the external benchmark BixBench, which is designed to assess agentic AI in bioinformatics tasks within a controlled Jupyter Notebook environment. We followed BixBench's official protocol by using its built-in agent, which operates through three primary commands: edit cell (to modify and execute code cells), list workdir (to inspect the workspace directory structure) and submit answer (to submit final responses). We configured BioMedAgent with GPT-4.1 as the base LLM and integrated the required data files into the Jupyter Notebook environment. The original BixBench questions were used as prompts to initiate data exploration and answer generation. The evaluation was conducted in three parallel testing groups: 296 open questions, 296 MCQs with refusal and 296 MCQs without refusal.

Tool-aware agent framework

Learning to use domain-specific tools. BioMedAgent allows users to add specialized tools into its localized workspace, thereby enabling effective use for various tasks. By learning tool documentation and preparing dependencies, tools can be integrated into the BioMedAgent's local workspace.

Tool documentation. The documentation includes six parts: name, identifier, description, input, output and example. The name specifies the tool's title. The tool identifier corresponds to the function name in the code and must remain consistent so that the LLM can be aware of the tool's presence within its workspace. The description provides an overview of the tool's functionality. The input and output cover details such as parameter names, types and descriptions. In addition, each documentation includes a simple contextual example of how to invoke the tool, helping BioMedAgent better understand its proper usage.

Containerization and central API. To manage the diverse implementation languages and environmental dependencies of different tools,

Docker containerization technology is used to create isolated execution environments for each tool. Python is then used to create a central API that acts as the main controller for initiating the program. This API handles input processing, launching the container, executing the command and retrieving results from a predefined workspace. This design ensures consistent, reproducible execution across heterogeneous tool environments.

Two-round tool retrieval via Tool Manager agent. A user's initial task request typically includes the user's available resources and their expected goals G, but often lacks specific details about the analytical process. For a given local tools set $T = \{t1, t2, t3\}$ in BioMedAgent's workspace, the Tool Manager evaluates the relevance of each tool using the user's initial description P and the tool documentation $D = \{d1, d2, d3\}$. This process generates a relevance score ranging from 0 to 1, reflecting how suitable each tool is, and forms the initial relevance score set $S = \{s1, s2, s3\}$. Tools with high relevance scores are selected for further exploration and usage recommendation.

To balance tool retrieval efficiency with context usage, we allowed users to specify a cut-off on the initial relevance scores. A higher cut-off improves tool retrieval precision but narrows recall and reduces the token cost of subsequent reasoning rounds (Supplementary Fig. 1a). Using 184 BioMed-AQA tasks that were successfully executed with LTU, we treated the tools actually invoked in these tasks as reference tools and evaluated different cut-offs for their ability to retain needed tools while limiting token cost. A cut-off of 0.6 provided the most favourable balance between recalling reference tools and controlling token cost (Supplementary Fig. 1b). In practice, this cut-off can be flexibly adjusted within 0.4–0.8 according to workspace size and available context budget.

The selected high-scoring tools are then used to generate usage recommendations, including relevant documentation and their upstream and downstream dependencies. These recommendations act as workflow 'seeds' guiding subsequent workflow expansion. Based on these results, the Tool Manager performs a second round of relevance assessment to refine the tool list. In this step, outlier tools with excessively high scores may be removed, while initially low-scoring but essential intermediate tools can be reintroduced by 'seeds' guiding. This two-round strategy helps BioMedAgent overcome common LLM limitations related to token capacity and long-context reasoning, ensuring that the final workflow includes the most necessary tools (Supplementary Fig. 2a).

Combining LTU and CTC. Compared to existing multi-omics agents like autoBA and general-purpose frameworks like LangGraph, which are often constrained by limited predefined tool usage, BioMedAgent provides much greater flexibility by combining LTU and CTC (Supplementary Fig. 2b). It integrates tools into the workspace on demand by learning their documentation and dependencies, allowing the agent to dynamically bind and invoke tools at runtime without relying on fixed configurations. When a required tool is missing, the agent can generate CTC as an alternative to complete the task.

LTU is ideal for integrating well-established domain-specific tools with specialized functionalities (for example, alignment, variant calling, cell clustering), which often involve complex dependencies. In fast-evolving omics research, these tools often come with complex environments and set-up procedures. We did not want BioMedAgent to be stuck solving set-up issues instead of doing the actual analysis, so we discouraged it from creating its own codes from scratch for such tasks. When specialized tools are available for the core analytical steps, and BioMedAgent only needs to handle lightweight tasks like filtering, splitting or result aggregation, the whole system becomes more stable, reproducible and suitable for real-world biomedical applications.

CTC works well for lightweight but essential operations such as scientific reasoning, summarizing, data filtering, splitting and aggregation, by using some publicly available packages already learned by LLMs. In cases where a required tool is missing, the agent attempts to generate its own CTC as an alternative to fill the gap.

This dual mechanism makes BioMedAgent both powerful and flexible, enabling it to support professional biomedical workflows while remaining adaptable and extensible across diverse research scenarios.

Inter-agent communication and self-evolving

To strengthen collaboration between agents, we introduced inter-agent communication and memory retention mechanisms through our IE and MR algorithms. IE allows agents, such as the planning agent and coding agent, or the coding agent and execution agent, to exchange feedback and context across steps, enabling dynamic adaptation and higher task success rates. MR assigns dedicated memory spaces to each agent (planning, coding and tool agents) and uses retrieval-augmented techniques to help agents reuse past experiences, making the system self-evolving.

Automatic code repair by the IE algorithm. BioMedAgent divides the workflow into multiple steps, transforming each step into a subtask. The Programmer agent then writes code for these subtasks using existing tool libraries and generates unit test code to verify correctness. The Executor agent runs this test code, handling variables, files and resources while producing the corresponding outputs. A shared resource pool maintains user inputs and intermediate outputs, ensuring that each step can access the necessary data. This structured approach ensures reliable and efficient code generation in multistep analytical tasks.

Bugs and errors are inevitable in the initial programming, and even experienced human programmers must iteratively refine and improve their code to achieve correct execution. To address this challenge, BioMedAgent uses a robust IE strategy for automated code repair. By combining forward information propagation and backward feedback, the IE algorithm enables BioMedAgent to broadcast localized issues to multiple upstream and downstream agents, transforming the problem-solving process from individual correction to collaborative intelligence.

When a unit test fails, the Executor agent collects detailed error information and forwards it to the Programmer agent group for exploration and discussion. This group examines the errors across dimensions such as syntax, parameters, logic and formatting, identifying the main cause and proposing corrective actions (Supplementary Fig. 2c). Empirically, IE effectively facilitates collaborative problem-solving, enabling multiple agents to surpass individual limitations by interacting and exploring together. It has successfully resolved most logical errors and code vulnerabilities introduced in the planning and coding phase. However, some issues may exceed the capabilities of LLMs, and repeated IE attempts might still lead to errors. To address this, we set an upper limit of three interaction rounds for IE during the planning and coding phases.

Tool, planning and coding memory in the MR algorithm. Memory helps organisms adapt to their environment, improve their behaviour, and learn socially as they evolve. Similarly, in intelligent agents, memory is essential for improving individual skills and ensuring effective communications[71]. In BioMedAgent, we include experiences from each problem we analyse and solve into three types of memories: tool memory, planning memory and coding memory.

Tool memory. When BioMedAgent first encounters a problem, it checks which tools are relevant in its workspace and figures out how to use them to solve the problem. Initially, it concentrates on each tool individually rather than how the tools work together. Once finishing the analysis and delivering the final result, BioMedAgent records each tool's contribution to the workflow and keeps this information in its memory.

Planning memory. While tool memory improves how each tool works, planning memory increases BioMedAgent's overall effectiveness. After completing an analysis, BioMedAgent simplifies the workflow by replacing specific filenames and unique identifiers with general placeholders, storing this simplified version in memory. This makes the workflow useful for more types of similar problems.

Coding memory. BioMedAgent also draws on previous coding experiences for similar programming tasks. This approach enables code reuse, reducing the cost of creating new code. After completing an analysis, BioMedAgent sets up key-value pairs that connect the goal to be achieved in each step to its specific code implementation. These pairs are saved in memory, allowing tasks to be matched based on their descriptions rather than exact code comparison, which is often more complex.

CMA and IMF strategy for memory update. In the self-evolving learning process, each learning phase is considered a 'round'. Within a round, experiences are accumulated independently, while memory updates occur between rounds. For BioMedAgent, we established two strategies for updating memory: CMA and IMF.

CMA. To maximize the preservation of group intelligence, this strategy retains all memories generated within a round without filtering or forgetting. This allows BioMedAgent to maintain a comprehensive memory base, storing all past experiences, which results in a large memory size.

IMF. Inspired by human memory's natural forgetting process, this strategy compares the solutions used for similar problems across different rounds. It retains the path with fewer steps and discards the others, treating the simpler steps as more efficient. This approach ensures that only one memory is kept for each type of problem, providing a richer variety of memories within the same memory space.

Reproducible, scalable and shareable intelligence with BioMedAgent

BioMedAgent addresses key challenges in biomedical data analysis by optimizing tool management, workflow planning and execution. It facilitates natural-language-based task initiation, enabling direct participation from biomedical professionals without specialized computational or bioinformatics training. The IE and MR framework allows BioMedAgent to accumulate extensive experience from tasks submitted by various users, all stored in natural language. This approach, centred on natural language for initiating tasks and storing experiences, enhances accessibility and scalability. By enabling users to create, refine, and share tools and insights through an intuitive, language-driven interface, BioMedAgent would help build a biomedical data analysis community around specific topics (Supplementary Fig. 3).

BioMedAgent could be deployed locally using online backends such as GPT-4/4o/4omini, Claude3 and local open-source LLMs such as Qwen3-235b-a22b. It offers an intuitive chat interface that simplifies initiating tasks, monitoring progress, and accessing results, making complex workflows more manageable and efficient on a local scale (Supplementary Fig. 3). This local deployment strategy improves data security and supports the development of specialized biomedical analysis systems. By customizing computational tools and accumulating experiences, BioMedAgent effectively adapts to the unique needs of specific tools and research domains, driving continuous advancements and fostering deeper domain-specific insights.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The BioMed-AQA benchmark (n = 327 open questions) introduced in this study is publicly available on Hugging Face at https://huggingface.co/datasets/BOBQWERA/biomed-aqa-dataset, which includes the full set of natural-language questions, reference steps and milestones for evaluation. The complementary BioMed-AQA-MCQ subset (n = 172 MCQs) is available at https://huggingface.co/datasets/BOBQWERA/biomed-mcq-dataset. The corresponding benchmark-related input data and milestone reference files are available via Zenodo at https://doi.org/10.5281/zenodo.17430550 (ref. 72). Evaluation testing of BioMedAgent for rounds = 0, 1, 2 under CMA and IMF on the BioMed-AQA benchmark, along with the interactive chat details that show the full process of task planning, execution and summarization for each question, are available at http://biomed.drai.cn.

Code availability

The open-source implementation of BioMedAgent, including its multi-agent framework and autoscoring evaluation agent, is available on GitHub at https://github.com/BOBQWERA/BioMedAgent.

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Author contributions

D.B. collected and interpreted the data and drafted the manuscript and figures. J.S. implemented and evaluated the algorithm and developed the software. K.L. analysed the data, drafted the figures and contributed to algorithm application. Z.H., W.H., J.H., S.Z. and S.L. participated in data processing and algorithm evaluation. P.H., Z.W. and S.W. contributed to tool preparation and website construction. T.W., K.G. and Y.W. assisted with literature collection and interpreted the results. L.Z., K.W., G.L., H.S. and Y.J. interpreted the results and supported application design. K.Z. and R.C. interpreted the results, revised the manuscript and figures and supervised the project. Y.Z. conceived the study, collected the data, revised the manuscript and figures and supervised the project. All authors reviewed and approved the final version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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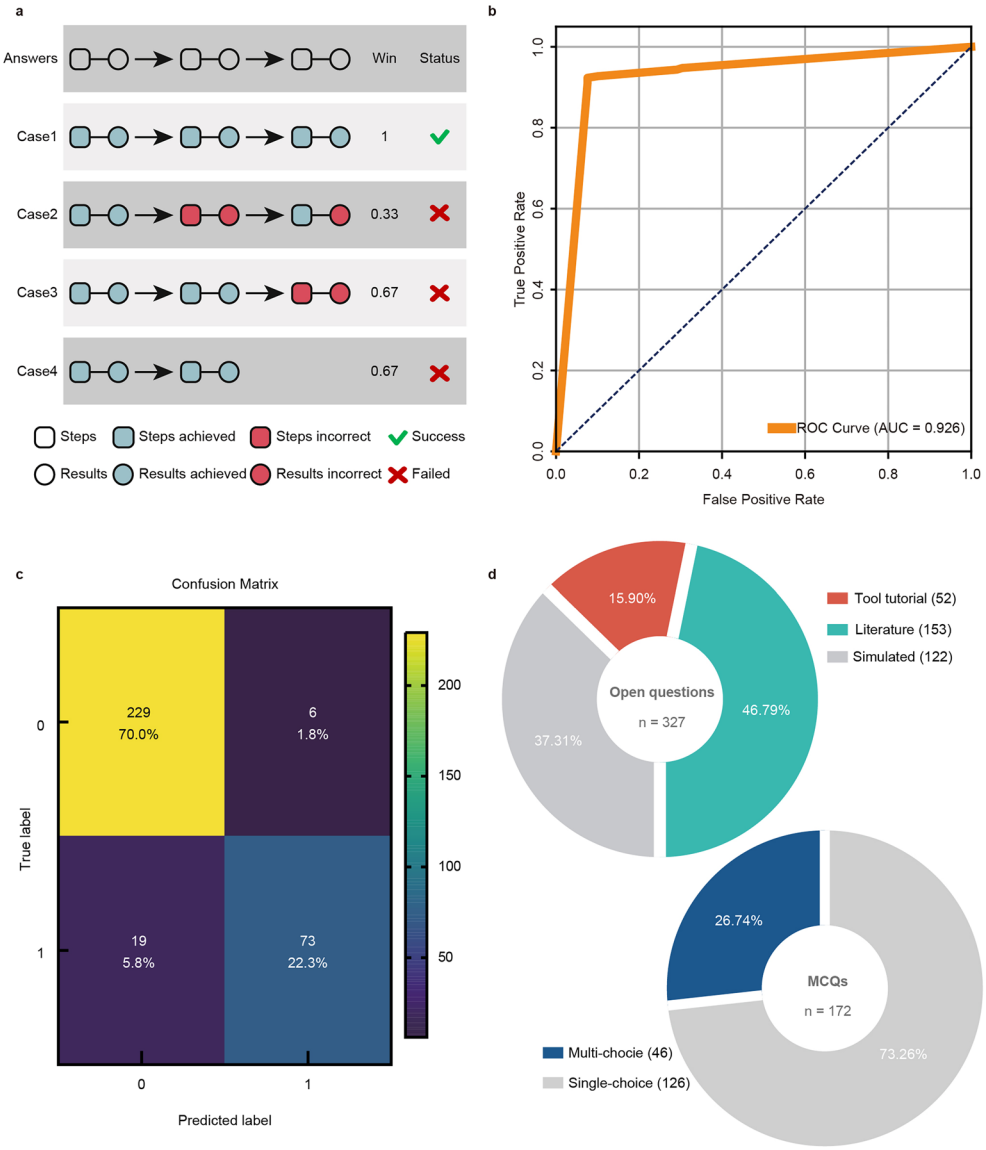
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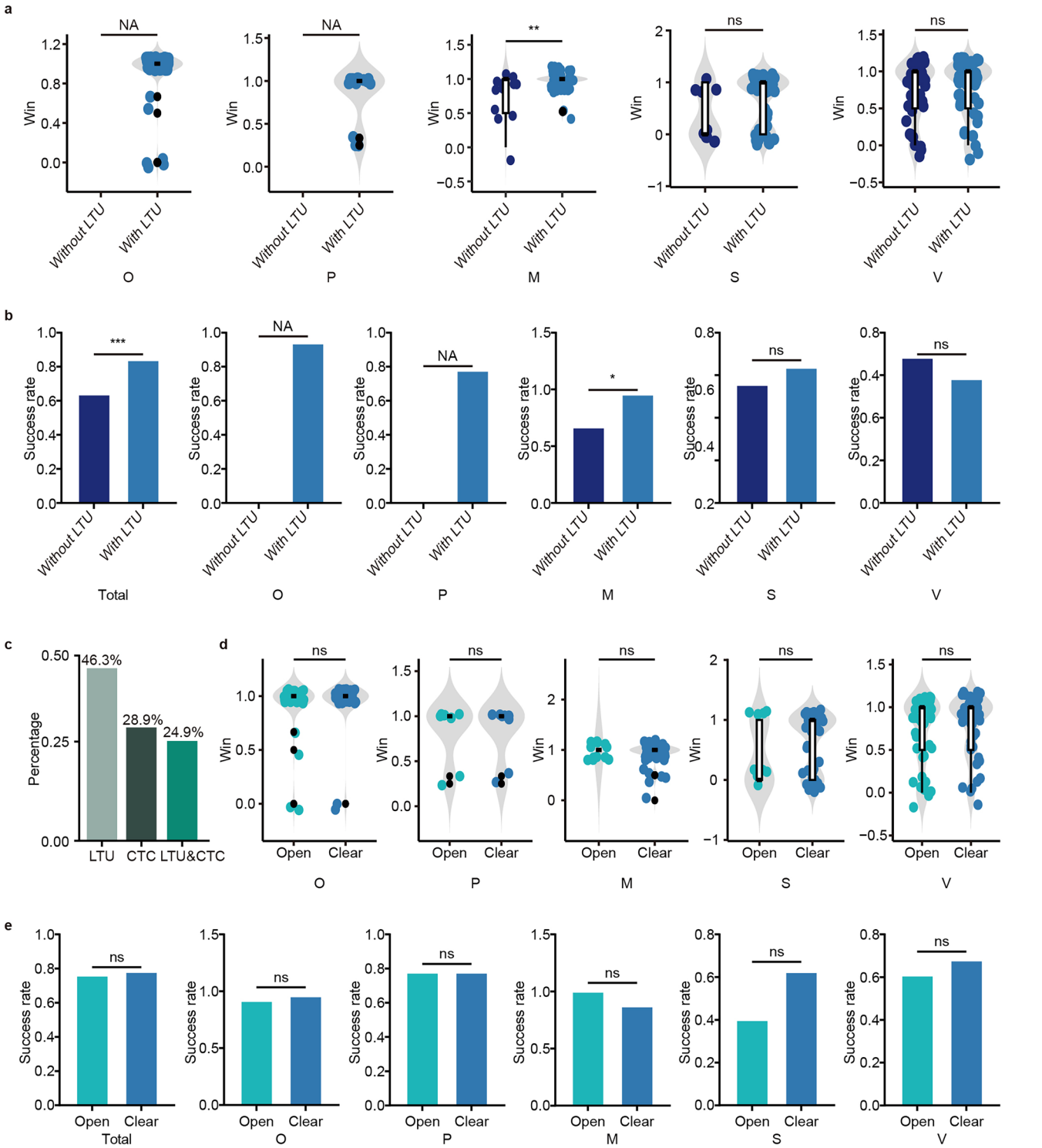
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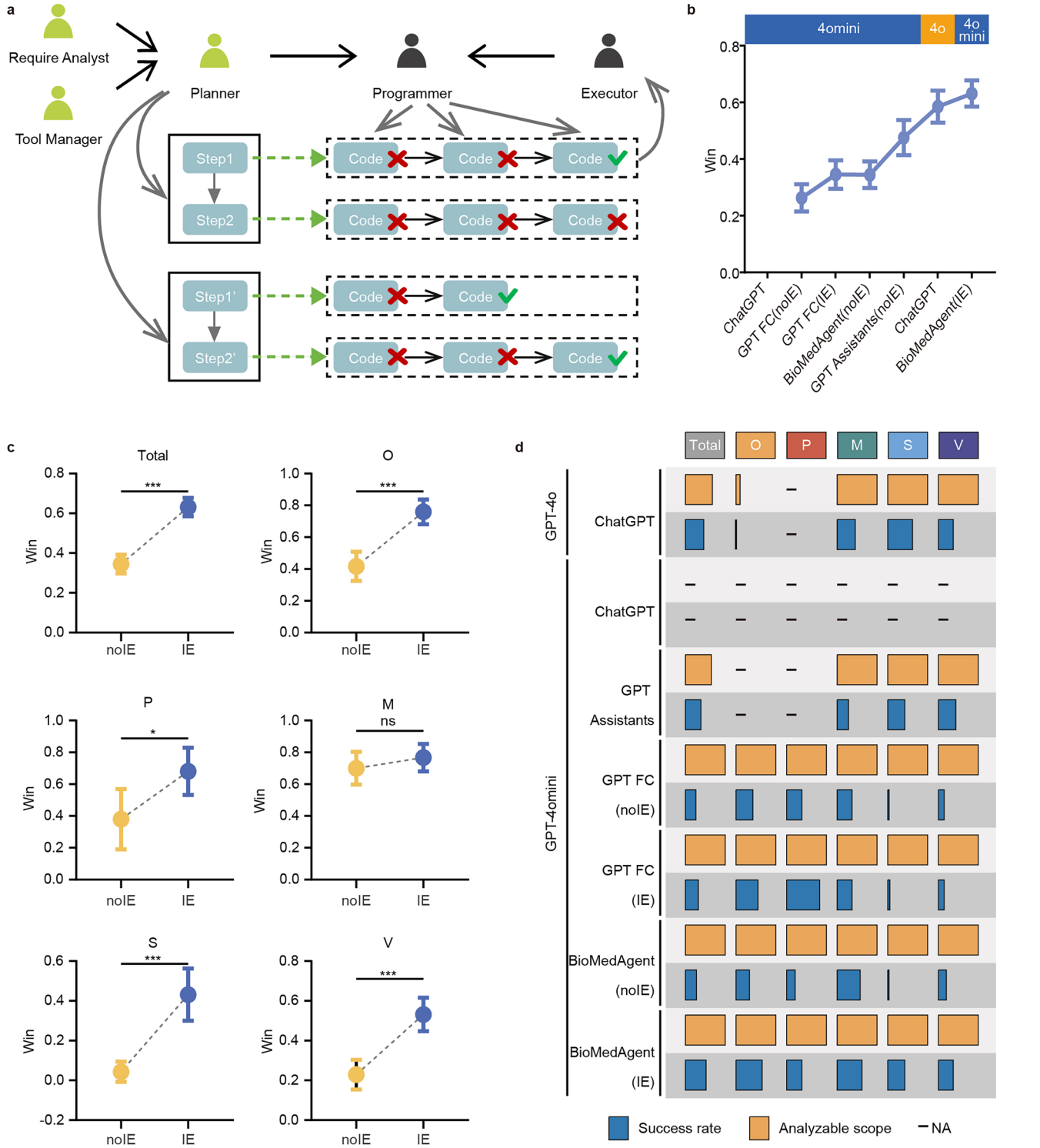
Extended Data Fig. 1 | Evaluation of BioMedAgent using BioMed-AQA. a. The calculation of the Win score and determining task execution status of success or failed. **b**. ROC curve demonstrating the performance of the autoscoring agent, showing high accuracy and reliability with an AUC of 0.926, indicating strong alignment with manual evaluations. **c**. Confusion matrix comparing autoscoring results with manual evaluations. **d**. Summary of BioMed-AQA and BioMed-AQA-MCQ. BioMed-AQA (n = 327) consists of open questions derived from three sources: simulated datasets (37.31%), literature-derived datasets (46.79%), and tool tutorial datasets (15.90%). BioMed-AQA-MCQ (n = 172) is a multiple-choice subset of BioMed-AQA, consisting of single-choice questions (73.26%) and multi-choice questions (26.74%), designed to enable automated and objective evaluation. All tasks from O, P, S in BioMed-AQA were designed with one corresponding multiple-choice question. M and V tasks were not included, as they are less suitable for the multiple-choice format.

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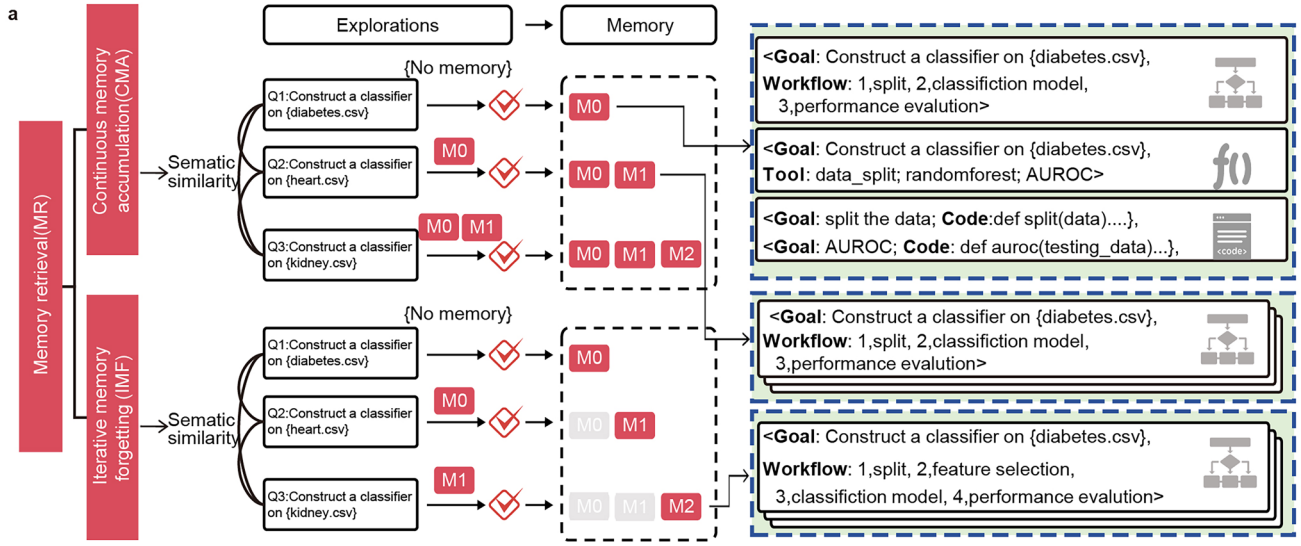
Extended Data Fig. 2 | Overall performance of BioMedAgent. a. Win score comparison between with and without LTU across O, P, M, S, V tasks, with p-values obtained via two-tailed t-test (p = 1.477e-03 for M, NA: Not Applicable, ns: p > 0.05, *: p < 0.05, **: p < 0.01, ***: p < 0.001). **b.** Success rate comparison between with and without LTU across Total, O, P, M, S, V tasks, with p-values obtained via two-sided chi-squared tests (p = 7.789e-05 for Total, p = 0.015 for M). **c.** Proportion of successful tasks on BioMed-AQA using only LTU, only CTC, and both. **d.** Win score comparison between open-step and clear-step questions across O, P, M, S, V tasks, with p-values obtained via two-tailed t-test. **e.** Success rate comparison between open-step and clear-step questions across Total, O, P, M, S, V tasks, with p-values obtained via two-sided chi-squared tests.

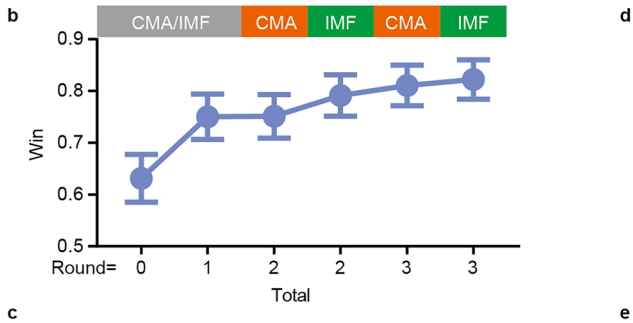
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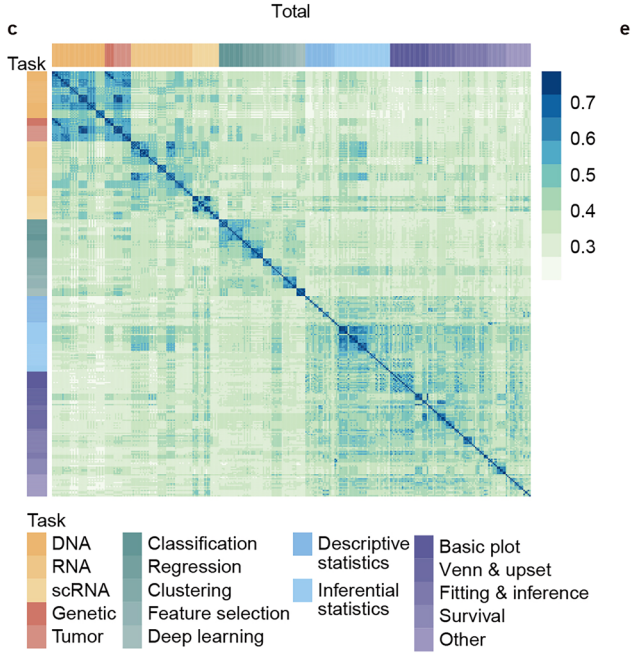


Extended Data Fig. 3 | Evaluation of the IE algorithm. a. Workflow of IE in planning and coding phases by Planner, Programmer and Executor agents. **b.** Win score of different LLM-based agents across Total tasks of BioMed-AQA (n = 327). Error bars represent the mean with 95% confidence intervals (CI). GPT FC stands for GPT Function Call. **c.** Win score comparison between noIE and IE across Total, O, P, M, S, V tasks, with p-values obtained via two-tailed paired t-test (Total: p = 1.091e-22, O: p = 4.236e-09, P: p = 0.019, M: p = 0.199, S: p = 5.118e-07, V: p = 5.530e-09, ns: p > 0.05, *: p < 0.05, **: p < 0.01, ***: p < 0.001). Error bars represent the mean with 95% CI. **d.** Analyzable scope and success rate of different LLM-based agents across Total, O, P, M, S, V tasks. The length of the bar represents the success rate or analyzable scope.

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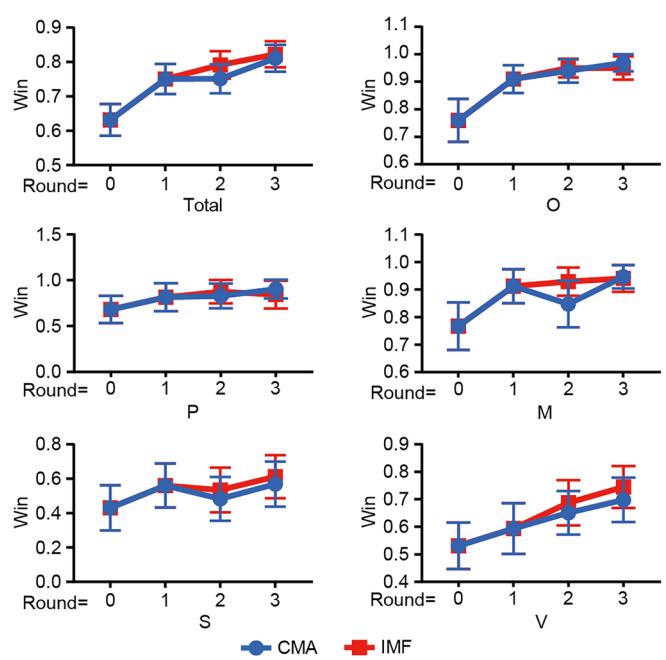






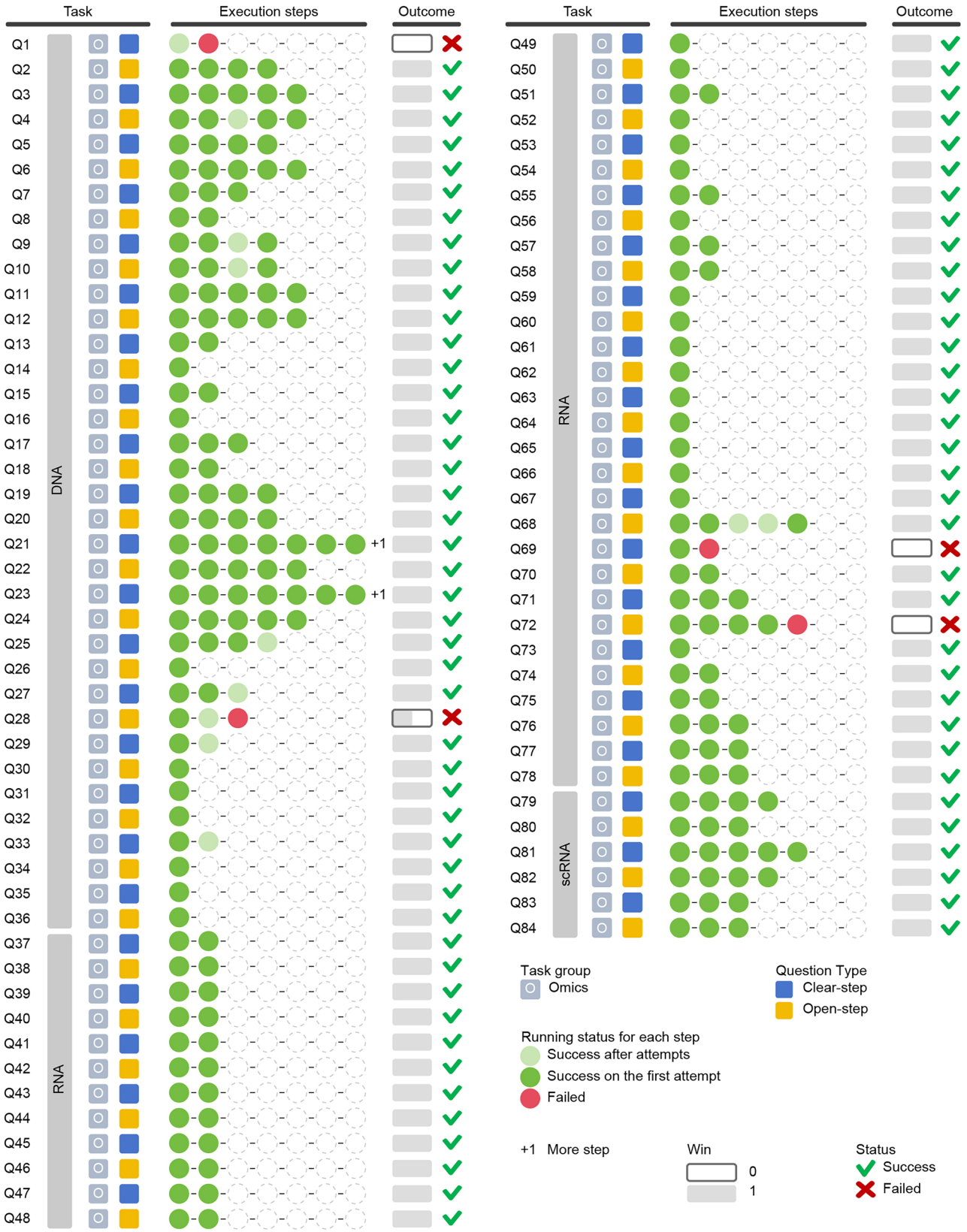
d

Model	Seen subset (n=229)	Unseen subset (n=98)
Before MR (round=0)	0.524	0.520
MR on seen (round=3)	0.755	0.694



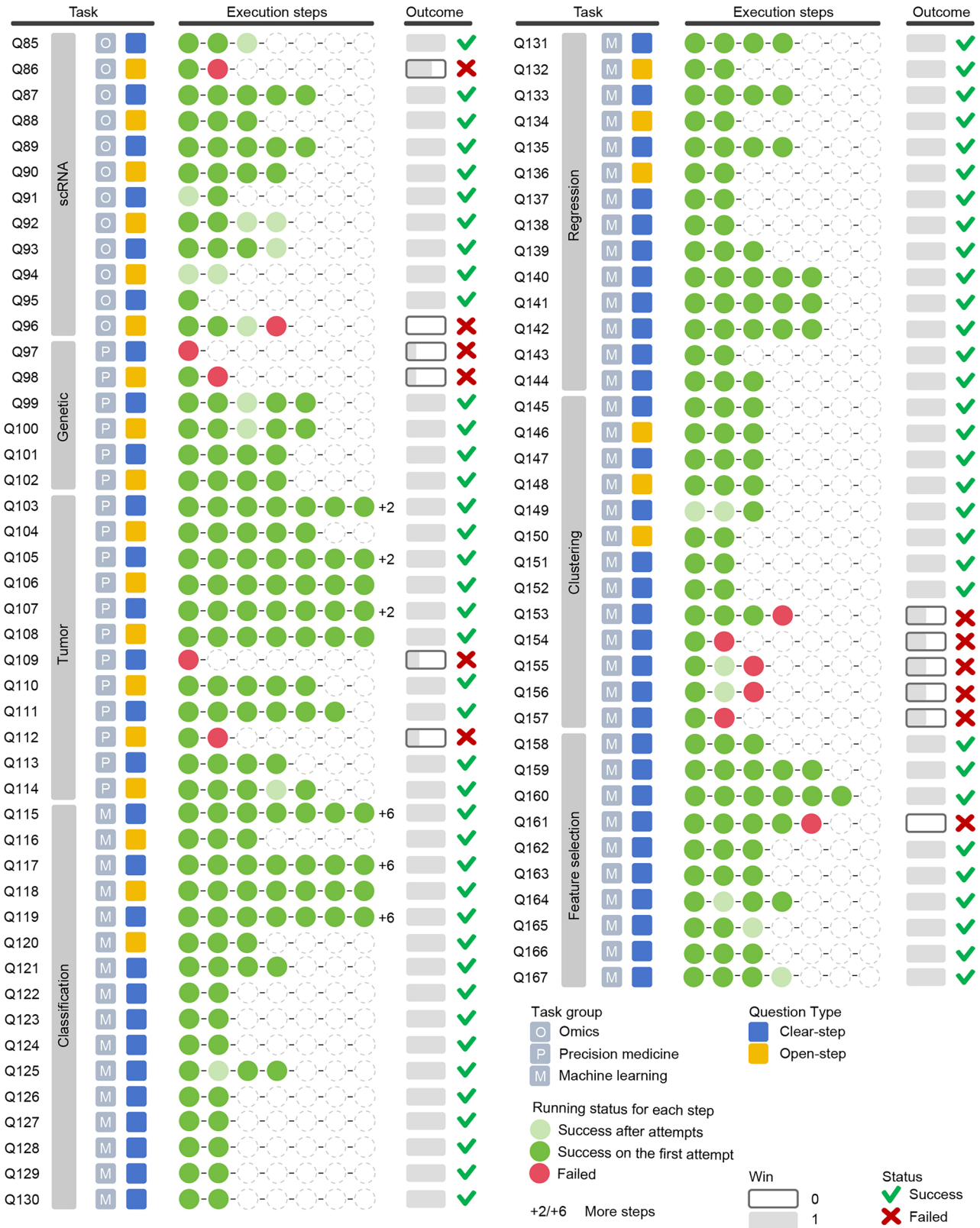
Extended Data Fig. 4 | Evaluation of the MR algorithm. a. Mechanisms of the CMA and IMF memory update strategies. M1 represents the new added memory in the learning of round 1, while M2 corresponds to round 2. **b.** Win scores under CMA and IMF memory update strategies across Total tasks (n = 327) in three rounds of learning. Error bars represent the mean with 95% CI. **c.** Semantic similarity between n = 327 BioMed-AQA questions. **d.** Evaluation of MR algorithm performance on unseen questions. This table shows the success rates for two models: 1). Before MR (round = 0) with no learning, and 2). MR on seen (round = 3) with three rounds of learning on the seen subset. **e.** Win scores under CMA and IMF strategies across Total, O, P, M, S, V tasks. Error bars represent the mean with 95% CI.

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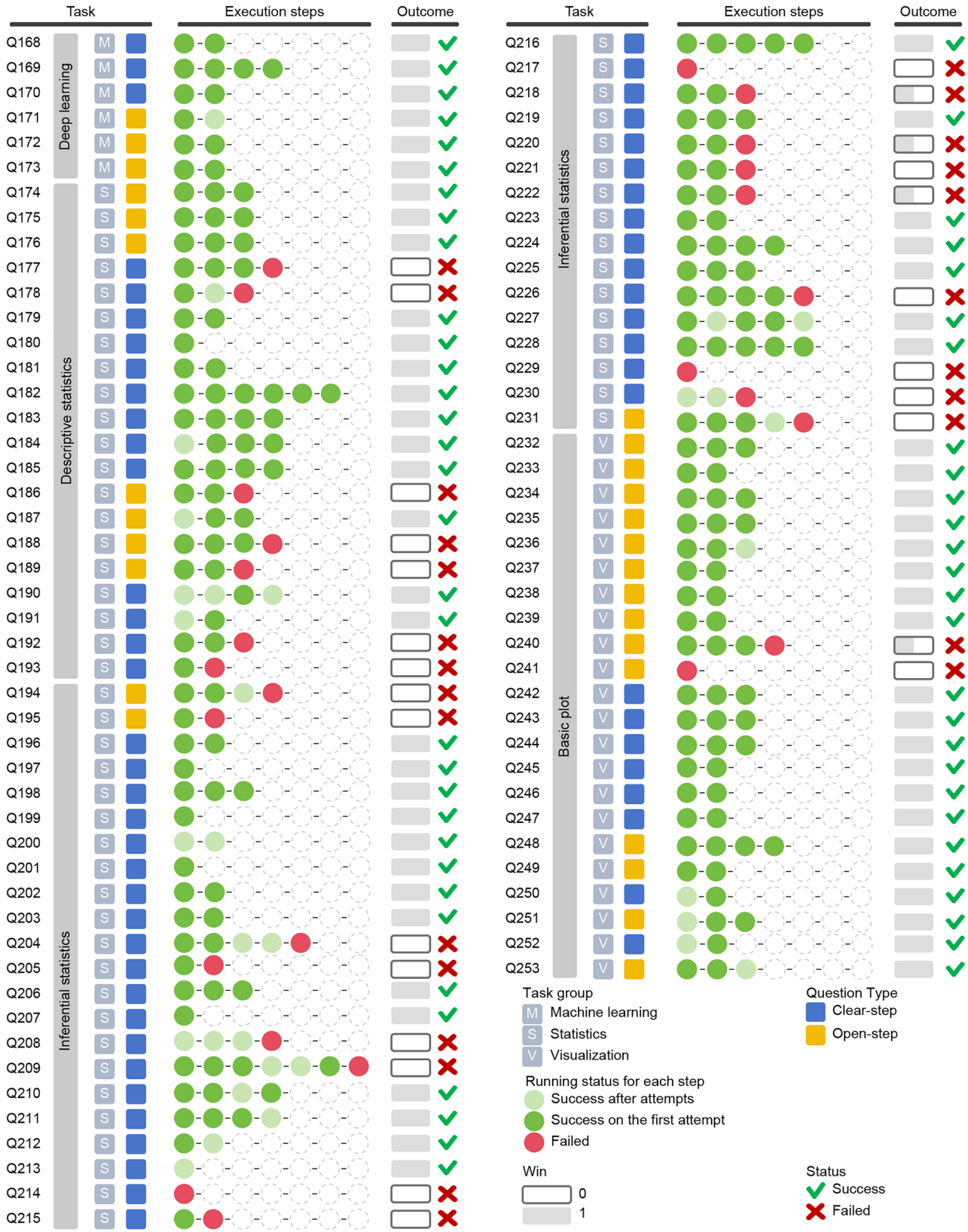
Extended Data Fig. 5 | Execution details of Q1-Q84 in BioMed-AQA. The detailed testing of each question in BioMed-AQA (Q1-Q84) by BioMedAgent, utilizing the IMF memory update strategy after three rounds of learning, includes the planned steps, Win scores, and execution outcomes. The planned steps are automatically generated by BioMedAgent, with +1 indicating one more step not shown.

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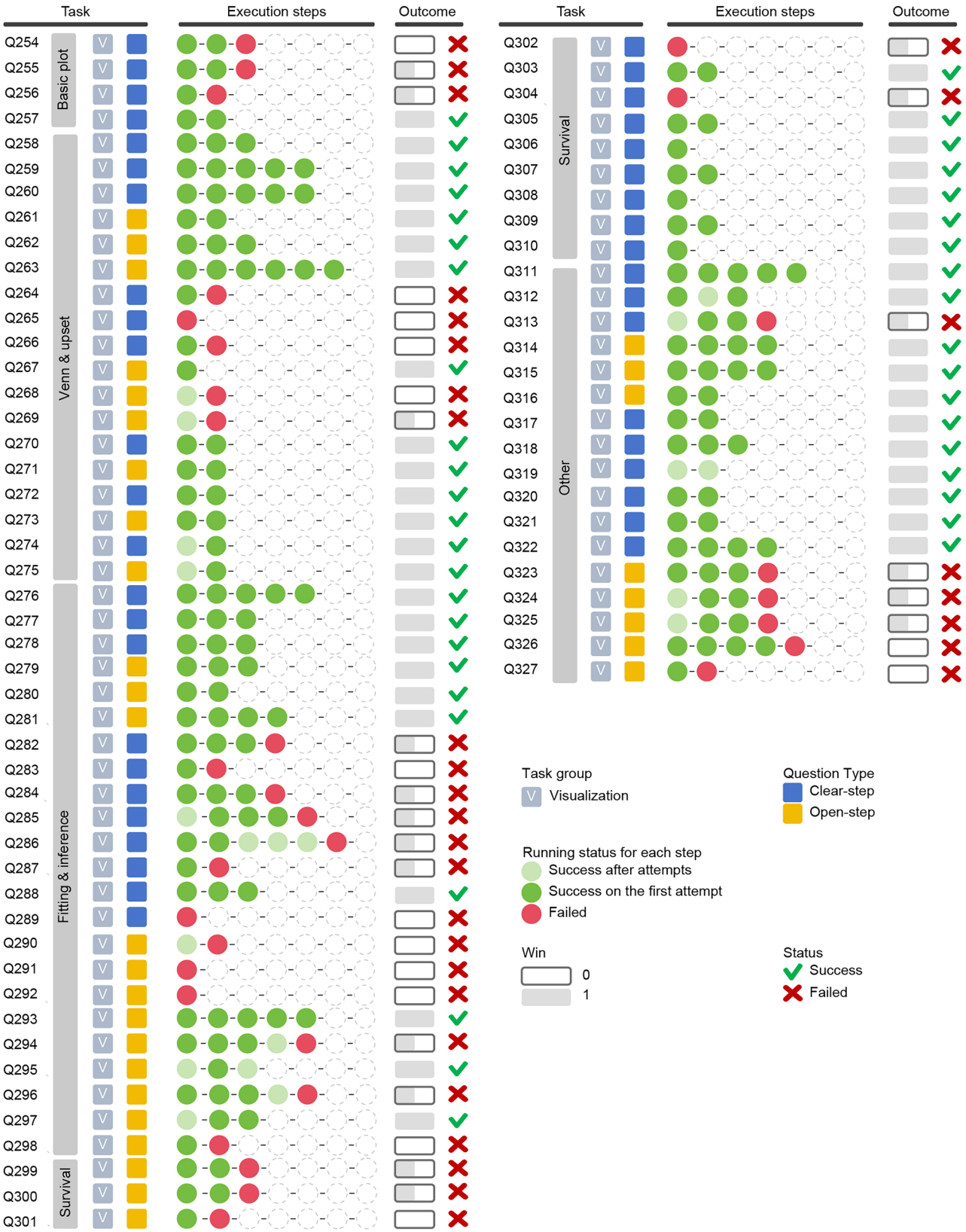
Extended Data Fig. 6 | Execution details of Q85-Q167 in BioMed-AQA. The detailed testing of each question in BioMed-AQA (Q85-Q167) by BioMedAgent, utilizing the IMF memory update strategy after three rounds of learning, includes the planned steps, Win scores, and execution outcomes. The planned steps are automatically generated by BioMedAgent, with +2/+6 indicates 2 or 6 more steps not shown.

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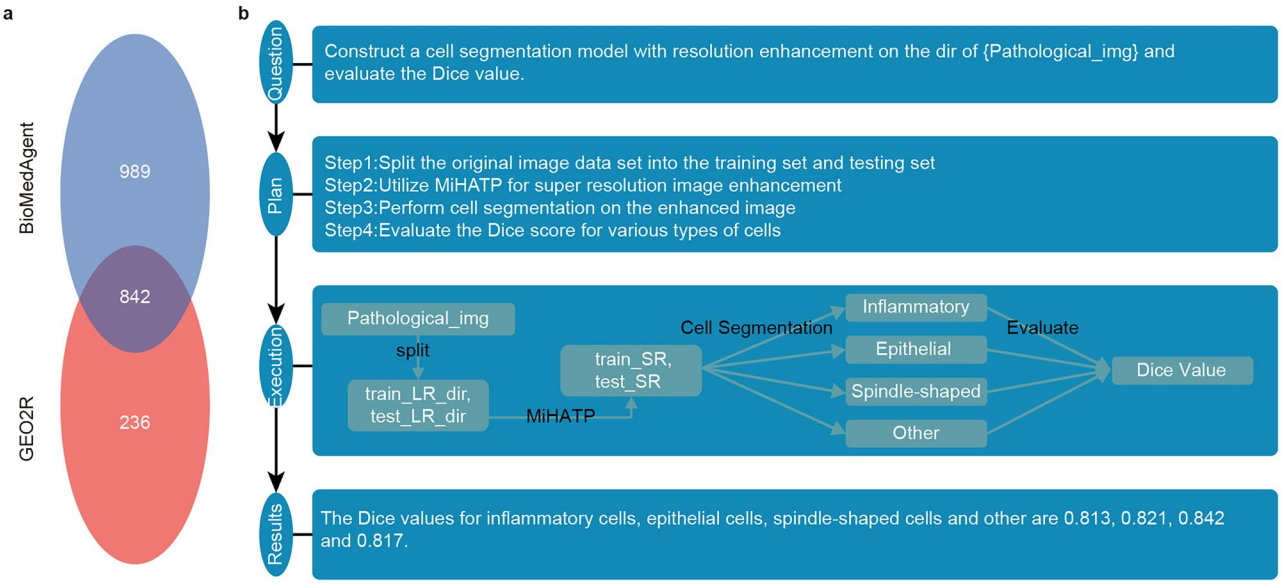
Extended Data Fig. 7 | Execution details of Q168-Q253 in BioMed-AQA. The detailed testing of each question in BioMed-AQA (Q168-Q253) by BioMedAgent, utilizing the IMF memory update strategy after three rounds of learning, includes the planned steps, Win scores, and execution outcomes. The planned steps are automatically generated by BioMedAgent.

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Extended Data Fig. 8 | Execution details of Q254-Q327 in BioMed-AQA. The detailed testing of each question in BioMed-AQA (Q254-Q327) by BioMedAgent, utilizing the IMF memory update strategy after three rounds of learning, includes the planned steps, Win scores, and execution outcomes. The planned steps are automatically generated by BioMedAgent.

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Extended Data Fig. 9 | Application examples of BioMedAgent. a. Comparison of DEGs identified by BioMedAgent and official online tool GEO2R. **b**. Construction of an automated workflow using BioMedAgent to perform cell segmentation with resolution enhancement.

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14 :20 You

I have a dataset {diabetes.csv}. The "label" column in the dataset is the classification label. Construct classification models and select the model with the best performance.

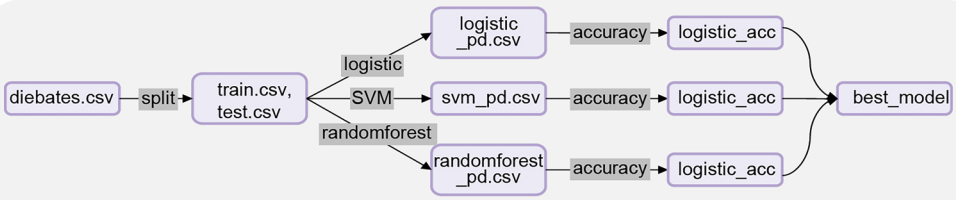
Require Analyst 14 :20

The task involved loading a diabetes dataset, preprocessing the data, building three different classification models, evaluating their performance using cross-validation, and selecting the best model based on the evaluation metrics.

Tool manager 14 :21

The top ranked tools are listed:
1. Random forest
2. SVM
3. Logistic
4. Accuracy evaluation
5. Data split

Workflow Planner 14 :22



Programmer 14 :25

```
1 from tool import *
2 import os
3
4 def code3():
5     # Calculate accuracy for Logistic Regression predictions
6     accuracy_logistic = accuracy_evaluation( "test.csv","label","Predicted")
7     # Calculate accuracy for SVM predictions
8     accuracy_svm = accuracy_calculate_tool("svm_predictions.csv","Target","Predicted")
```

Executor 14 :30

Invoke the code3 program to run.
Code3 failed, attempting to automatically update parameters.
1 attempt after code3 failure.
2 attempt after code3 failure.
3 attempt after code3 failure.
Code3 ran successfully.

Reporter 14 :32

The SVM model is selected as the best classifier for the diabetes dataset based on its superior performance in the evaluation. This model is recommended for further development and deployment in related tasks due to its robustness and effectiveness in handling the complexities of the dataset.
View report.

Extended Data Fig. 10 | Intuitive chat interface of BioMedAgent. This figure illustrates the interactive chat interface of BioMedAgent, where multiple agents collaborate on task planning, execution, and results aggregation. Users interact with the system in natural language to initiate and monitor the execution of workflows, including model selection and evaluation in the demo example. The system autonomously plans, executes, and summarizes the results.

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Reporting Summary

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
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☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
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☒	☐ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	The literature- and tool-tutorial-based data used to construct the BioMed-AQA benchmark were collected from publicly available datasets or resources, with no software required for data collection. The simulated data of BioMed-AQA was generated using the open-source tool VarBen (https://github.com/nccl-jmli/VarBen).
Data analysis	We used our newly developed BioMedAgent to perform analysis on the collected data. BioMedAgent is open-source and available at https://github.com/BOBQWERA/BioMedAgent.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

The BioMed-AQA benchmark (n=327 open questions) introduced in this study is publicly available on Hugging Face at https://huggingface.co/datasets/BOBQWERA/

biomed-aqa-dataset, which includes the full set of natural language questions, reference steps, and milestones for evaluation. The complementary BioMed-AQA-MCQ subset (n=172 multiple-choice questions) is available at https://huggingface.co/datasets/BOBQWERA/biomed-mcq-dataset. The corresponding benchmark-related input data and milestone reference files have been deposited in Zenodo at https://doi.org/10.5281/zenodo.17430550. Evaluation testing of BioMedAgent for rounds = 0, 1, 2 under CMA and IMF on the BioMed-AQA benchmark, along with the interactive chat details that demonstrate the full process of task planning, execution, and summarization for each question, are accessible at http://biomed.drai.cn.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	No sex or gender information were collected or analyzed, as this study focused on developing an autonomous agent framework for data analysis.
Reporting on race, ethnicity, or other socially relevant groupings	No race, ethnicity, or other socially relevant groupings were collected or analyzed in this study.
Population characteristics	No population characteristics were collected or analyzed in this study.
Recruitment	No participants were recruited, as all analyses were conducted on publicly available, de-identified data.
Ethics oversight	Ethics approval was not required because all analyses were conducted on publicly available, de-identified data.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The sample consisted of 327 biomedical data analysis questions across five categories (O, P, M, S, and V). No formal statistical sample size calculation was performed; the size was selected to capture task diversity and enable a comprehensive evaluation of BioMedAgent.
Data exclusions	No data were excluded from the analyses. All 327 questions were included in the evaluation to assess BioMedAgent's performance.
Replication	No replication was performed, as the work is a method evaluation on predefined data analysis questions rather than an experiment with cohorts or participants.
Randomization	Randomization is not applicable to this study as it focuses on evaluating BioMedAgent's performance across a set of predefined biomedical analysis questions rather than involving experimental groups or participants. The 327 questions were selected based on their relevance to various biomedical data analysis tasks, without the need for randomization.
Blinding	Blinding is not applicable to our study as all the data were de-identified.

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☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	No plants related.
Novel plant genotypes	No plants related.
Authentication	No plants related.